2.14 Hyperbolic Functions: Osborn's Rule & Proof

Worked Solutions

1 a)

$$\sin^2 x + \cos^2 x \equiv 1 \Rightarrow -\sinh^2 x + \cosh^2 x \equiv 1$$

Therefore,

$$\cosh^2 x - \sinh^2 x \equiv 1.$$

1 b)

$$\begin{aligned}
1 + \cot^2 x &\equiv \operatorname{cosec}^2 x \\
1 + \frac{\cos^2 x}{\sin^2 x} &\equiv \frac{1}{\sin^2 x} \\
1 + \frac{\cosh^2 x}{-\sinh^2 x} &\equiv \frac{1}{-\sinh^2 x} \\
1 - \coth^2 x &\equiv -\operatorname{cosech}^2 x \\
\coth^2 x - 1 &\equiv \operatorname{cosech}^2 x.
\end{aligned}$$

1 c)

$$\begin{aligned}
\tan^2 x + 1 &\equiv \sec^2 x \\
\frac{\sin^2 x}{\cos^2 x} + 1 &\equiv \frac{1}{\cos^2 x} \\
\frac{-\sinh^2 x}{\cosh^2 x} + 1 &\equiv \frac{1}{\cosh^2 x} \\
-\tanh^2 x + 1 &\equiv \operatorname{sech}^2 x \\
1 - \tanh^2 x &\equiv \operatorname{sech}^2 x.
\end{aligned}$$

1 d)

$$\begin{aligned}
\sin(x+y) &\equiv \sin x \cos y + \cos x \sin y \\
\sinh(x+y) &\equiv \sinh x \cosh y + \cosh x \sinh y.
\end{aligned}$$

1 e)

$$\begin{aligned}
\sin(x-y) &\equiv \sin x \cos y - \cos x \sin y \\
\sinh(x-y) &\equiv \sinh x \cosh y - \cosh x \sinh y.
\end{aligned}$$

1 f)

$$\begin{aligned}\cos(x+y) &\equiv \cos x \cos y - \sin x \sin y \\ \cosh(x+y) &\equiv \cosh x \cosh y - (-\sinh x \sinh y) \\ &\equiv \cosh x \cosh y + \sinh x \sinh y.\end{aligned}$$

1 g)

$$\begin{aligned}\cos(x-y) &\equiv \cos x \cos y + \sin x \sin y \\ \cosh(x-y) &\equiv \cosh x \cosh y + (-\sinh x \sinh y) \\ &\equiv \cosh x \cosh y - \sinh x \sinh y.\end{aligned}$$

1 h)

$$\begin{aligned}\tan(x+y) &\equiv \frac{\tan x + \tan y}{1 - \tan x \tan y} \\ \frac{\sin(x+y)}{\cos(x+y)} &\equiv \frac{\frac{\sin x}{\cos x} + \frac{\sin y}{\cos y}}{1 - \frac{\sin x}{\cos x}\frac{\sin y}{\cos y}} \\ \frac{\sinh(x+y)}{\cosh(x+y)} &\equiv \frac{\frac{\sinh x}{\cosh x} + \frac{\sinh y}{\cosh y}}{1 + \frac{\sinh x \sinh y}{\cosh x \cosh y}} \\ \tanh(x+y) &\equiv \frac{\tanh x + \tanh y}{1 + \tanh x \tanh y}.\end{aligned}$$

1 i)

$$\begin{aligned}\tan(x-y) &\equiv \frac{\tan x - \tan y}{1 + \tan x \tan y} \\ \frac{\sin(x-y)}{\cos(x-y)} &\equiv \frac{\frac{\sin x}{\cos x} - \frac{\sin y}{\cos y}}{1 + \frac{\sin x}{\cos x}\frac{\sin y}{\cos y}} \\ \frac{\sinh(x-y)}{\cosh(x-y)} &\equiv \frac{\frac{\sinh x}{\cosh x} - \frac{\sinh y}{\cosh y}}{1 - \frac{\sinh x \sinh y}{\cosh x \cosh y}} \\ \tanh(x-y) &\equiv \frac{\tanh x - \tanh y}{1 - \tanh x \tanh y}.\end{aligned}$$

1 j)

$$\begin{aligned}\sin 2x &\equiv 2 \sin x \cos x \\ \sinh 2x &\equiv 2 \sinh x \cosh x.\end{aligned}$$

1 k)

$$\begin{aligned}\cos 2x &\equiv \cos^2 x - \sin^2 x \\ \cosh 2x &\equiv \cosh^2 x - (-\sinh^2 x) \\ &\equiv \cosh^2 x + \sinh^2 x.\end{aligned}$$

1 l)

$$\begin{aligned}\cos 2x &\equiv 2\cos^2 x - 1 \\ \cosh 2x &\equiv 2\cosh^2 x - 1.\end{aligned}$$

1 m)

$$\begin{aligned}\cos 2x &\equiv 1 - 2\sin^2 x \\ \cosh 2x &\equiv 1 - 2(-\sinh^2 x) \\ &\equiv 1 + 2\sinh^2 x.\end{aligned}$$

1 n)

$$\begin{aligned}\tan 2x &\equiv \frac{2\tan x}{1-\tan^2 x} \\ \frac{\sin 2x}{\cos 2x} &\equiv \frac{2\frac{\sin x}{\cos x}}{1-\frac{\sin^2 x}{\cos^2 x}} \\ \frac{\sinh 2x}{\cosh 2x} &\equiv \frac{2\frac{\sinh x}{\cosh x}}{1+\frac{\sinh^2 x}{\cosh^2 x}} \\ \tanh 2x &\equiv \frac{2\tanh x}{1+\tanh^2 x}.\end{aligned}$$

1 o)

$$\begin{aligned}\sin 3x &\equiv 3\sin x - 4\sin^3 x \\ &\equiv 3\sin x - 4\sin x \sin^2 x \\ \sinh 3x &\equiv 3\sinh x - 4\sinh x(-\sinh^2 x) \\ &\equiv 3\sinh x + 4\sinh^3 x.\end{aligned}$$

1 p)

$$\begin{aligned}\cos 3x &\equiv 4\cos^3 x - 3\cos x \\ \cosh 3x &\equiv 4\cosh^3 x - 3\cosh x.\end{aligned}$$

1 q)

$$\begin{aligned}
\tan 3x &\equiv \frac{3\tan x - \tan^3 x}{1 - 3\tan^2 x} \\
\frac{\sin 3x}{\cos 3x} &\equiv \frac{3\frac{\sin x}{\cos x} - \frac{\sin^3 x}{\cos^3 x}}{1 - 3\frac{\sin^2 x}{\cos^2 x}} \\
\frac{\sinh 3x}{\cosh 3x} &\equiv \frac{3\frac{\sinh x}{\cosh x} - \frac{\sinh x(-\sinh^2 x)}{\cosh^3 x}}{1 + 3\frac{\sinh^2 x}{\cosh^2 x}} \\
\tanh 3x &\equiv \frac{3\tanh x + \tanh^3 x}{1 + 3\tanh^2 x}.
\end{aligned}$$

1 r)

$$\begin{aligned}
\sin 4x &\equiv \cos x(4\sin x - 8\sin^3 x) \\
\sinh 4x &\equiv \cosh x(4\sinh x - 8\sinh x(-\sinh^2 x)) \\
&\equiv \cosh x(4\sinh x + 8\sinh^3 x).
\end{aligned}$$

1 s)

$$\begin{aligned}
\cos 4x &\equiv 8\cos^4 x - 8\cos^2 x + 1 \\
\cosh 4x &\equiv 8\cosh^4 x - 8\cosh^2 x + 1.
\end{aligned}$$

1 t)

$$\begin{aligned}
\tan 4x &\equiv \frac{4\tan x - 4\tan^3 x}{1 - 6\tan^2 x + \tan^4 x} \\
\frac{\sin 4x}{\cos 4x} &\equiv \frac{4\frac{\sin x}{\cos x} - 4\frac{\sin^3 x}{\cos^3 x}}{1 - 6\frac{\sin^2 x}{\cos^2 x} + \frac{\sin^4 x}{\cos^4 x}} \\
\frac{\sinh 4x}{\cosh 4x} &\equiv \frac{4\frac{\sinh x}{\cosh x} - 4\frac{\sinh x(-\sinh^2 x)}{\cosh^3 x}}{1 + 6\frac{\sinh^2 x}{\cosh^2 x} + \frac{(-\sinh^2 x)^2}{\cosh^4 x}} \\
\tanh 4x &\equiv \frac{4\tanh x + 4\tanh^3 x}{1 + 6\tanh^2 x + \tanh^4 x}.
\end{aligned}$$

2 a)

$$\begin{aligned}
\frac{\sinh x}{\cosh^2 x} &\equiv \frac{\sinh x}{\cosh x} \times \frac{1}{\cosh x} \\
&\equiv \tanh x \operatorname{sech} x.
\end{aligned}$$

2 b)

$$\begin{aligned}\frac{\cosh x}{\sinh^2 x} &\equiv \frac{\cosh x}{\sinh x} \times \frac{1}{\sinh x} \\ &\equiv \coth x \operatorname{cosech} x.\end{aligned}$$

3 a)

$$\begin{aligned}\operatorname{cosech} x \sinh x \cosh x &\equiv \frac{1}{\sinh x} \sinh x \cosh x \\ &\equiv \cosh x.\end{aligned}$$

3 b)

$$\begin{aligned}\tanh x \operatorname{sech} x \cosh x &\equiv \tanh x \times \frac{1}{\cosh x} \times \cosh x \\ &\equiv \tanh x.\end{aligned}$$

3 c)

$$\begin{aligned}\tanh x \operatorname{cosech} x \coth x \sinh x &\equiv \tanh x \times \frac{1}{\sinh x} \times \frac{1}{\tanh x} \times \sinh x \\ &\equiv 1.\end{aligned}$$

4 a)

$$\begin{aligned}\frac{\operatorname{sech} x}{\operatorname{cosech} x} &\equiv \frac{1/\cosh x}{1/\sinh x} \\ &\equiv \frac{\sinh x}{\cosh x} \equiv \tanh x.\end{aligned}$$

4 b)

$$\begin{aligned}\frac{\coth x}{\operatorname{sech} x} &\equiv \frac{\cosh x/\sinh x}{1/\cosh x} \\ &\equiv \frac{\cosh^2 x}{\sinh x} \equiv \cosh^2 x \operatorname{cosech} x.\end{aligned}$$

4 c)

$$\begin{aligned}\frac{\operatorname{cosech} x}{\tanh x} &\equiv \frac{1/\sinh x}{\sinh x/\cosh x} \\ &\equiv \frac{\cosh x}{\sinh^2 x} \equiv \cosh x \operatorname{cosech}^2 x.\end{aligned}$$

4 d)

$$\begin{aligned}\frac{\operatorname{sech} x}{\coth x} &\equiv \frac{1/\cosh x}{\cosh x/\sinh x} \\ &\equiv \frac{\sinh x}{\cosh^2 x} \equiv \sinh x \operatorname{sech}^2 x.\end{aligned}$$

4 e)

$$\begin{aligned}\frac{\operatorname{sech}^2 x}{\coth x} &\equiv \frac{1/\cosh^2 x}{\cosh x/\sinh x} \\ &\equiv \frac{\sinh x}{\cosh^3 x} \equiv \sinh x \operatorname{sech}^3 x.\end{aligned}$$

4 f)

$$\begin{aligned}\frac{\operatorname{cosech}^2 x \coth^2 x}{\operatorname{sech} x \sinh x} &\equiv \frac{\frac{1}{\sinh^2 x} \times \frac{\cosh^2 x}{\sinh^2 x}}{\frac{1}{\cosh x} \times \sinh x} \\ &\equiv \frac{\cosh^2 x}{\sinh^4 x} \times \frac{\sinh x}{\cosh x} \\ &\equiv \frac{\cosh^3 x}{\sinh^5 x} \equiv \cosh^3 x \operatorname{cosech}^5 x.\end{aligned}$$

5)

$$\begin{aligned}-\ln(k+\sqrt{k^2-1}) &= \ln((k+\sqrt{k^2-1})^{-1}) \\ &= \ln\left(\frac{1}{k+\sqrt{k^2-1}}\right) \\ &= \ln\left(\frac{1}{k+\sqrt{k^2-1}} \times \frac{k-\sqrt{k^2-1}}{k-\sqrt{k^2-1}}\right) \\ &= \ln\left(\frac{k-\sqrt{k^2-1}}{k^2-(k^2-1)}\right) \\ &= \ln\left(\frac{k-\sqrt{k^2-1}}{1}\right) \\ &= \ln(k-\sqrt{k^2-1}).\end{aligned}$$

6)

$$\begin{aligned}-\ln(k+\sqrt{k^2+1}) &= \ln((k+\sqrt{k^2+1})^{-1}) \\ &= \ln\left(\frac{1}{k+\sqrt{k^2+1}}\right)\end{aligned}$$

$$
\begin{aligned}
&= \ln\left(\frac{1}{k+\sqrt{k^2+1}} \times \frac{k-\sqrt{k^2+1}}{k-\sqrt{k^2+1}}\right) \\
&= \ln\left(\frac{k-\sqrt{k^2+1}}{k^2-(k^2+1)}\right) \\
&= \ln\left(\frac{k-\sqrt{k^2+1}}{-1}\right) \\
&= \ln(-k+\sqrt{k^2+1}) \\
&\not\equiv \ln(k-\sqrt{k^2+1}).
\end{aligned}
$$

7)

$$
\begin{aligned}
\cosh x &= k \\
\Rightarrow \frac{e^x+e^{-x}}{2} &= k \\
\Rightarrow e^x + e^{-x} &= 2k \\
\Rightarrow e^{2x} + 1 &= 2ke^x \\
\Rightarrow e^{2x} - 2ke^x + 1 &= 0 \\
\Rightarrow (e^x-k)^2 - k^2 + 1 &= 0 \\
\Rightarrow (e^x-k)^2 &= k^2-1 \\
\Rightarrow e^x - k &= \pm\sqrt{k^2-1} \\
\Rightarrow e^x &= k \pm \sqrt{k^2-1} \\
\Rightarrow x &= \ln(k \pm \sqrt{k^2-1}).
\end{aligned}
$$

8)

$$
\begin{aligned}
\cosh x = 4 \Rightarrow \frac{e^x+e^{-x}}{2} &= 4 \\
\Rightarrow e^{2x} - 8e^x + 1 &= 0 \\
\Rightarrow (e^x-4)^2 - 16 + 1 &= 0 \\
\Rightarrow (e^x-4)^2 &= 15 \\
\Rightarrow e^x &= 4 \pm \sqrt{15} \\
\Rightarrow x &= \ln(4 \pm \sqrt{15}).
\end{aligned}
$$

Now consider

$$
\begin{aligned}
\ln(4-\sqrt{15}) &= \ln\left((4-\sqrt{15})\frac{4+\sqrt{15}}{4+\sqrt{15}}\right) \\
&= \ln\left(\frac{16-15}{4+\sqrt{15}}\right) = \ln\left(\frac{1}{4+\sqrt{15}}\right) \\
&= \ln((4+\sqrt{15})^{-1}) = -\ln(4+\sqrt{15}).
\end{aligned}
$$

So $\ln(4-\sqrt{15}) = -\ln(4+\sqrt{15})$. The solutions of $\cosh x = 4$ are $x = \ln(4 \pm \sqrt{15})$, which can also be written as $x = \pm\ln(4+\sqrt{15})$.

9 a)

$$\begin{aligned} 8\cosh^2 x + 7\sinh^2 x &\equiv 8\cosh^2 x + 7(\cosh^2 x - 1) \\ &\equiv 15\cosh^2 x - 7. \end{aligned}$$

9 b)

$$\begin{aligned} 3\sinh^2 x - 13\cosh^2 x &\equiv 3(\cosh^2 x - 1) - 13\cosh^2 x \\ &\equiv -10\cosh^2 x - 3. \end{aligned}$$

10)

$$\begin{aligned} 3\cosh^2 x - \sinh^2 x &= 5 \\ 3\cosh^2 x - (\cosh^2 x - 1) &= 5 \\ 2\cosh^2 x + 1 &= 5 \\ \cosh^2 x &= 2 \\ \cosh x &= \pm\sqrt{2}. \end{aligned}$$

$\cosh x = -\sqrt{2}$ has no real solutions. If $\cosh x = \sqrt{2}$, then

$$x = \pm \ln\left(\sqrt{2} + \sqrt{(\sqrt{2})^2 - 1}\right) = \pm \ln(\sqrt{2} + 1).$$

So $x = \ln(1 + \sqrt{2})$ or $x = \ln(-1 + \sqrt{2})$.

11 a)

$$\begin{aligned} \cosh^2 x - \sinh x - 7 &= 0 \\ (\sinh^2 x + 1) - \sinh x - 7 &= 0 \\ \sinh^2 x - \sinh x - 6 &= 0 \\ (\sinh x - 3)(\sinh x + 2) &= 0 \\ \sinh x = 3 \text{ or } \sinh x &= -2. \end{aligned}$$

If $\sinh x = 3$, then $x = \ln(3 + \sqrt{3^2 + 1}) = \ln(3 + \sqrt{10})$. If $\sinh x = -2$, then $x = \ln(-2 + \sqrt{(-2)^2 + 1}) = \ln(-2 + \sqrt{5})$. So $x = \ln(3 + \sqrt{10})$ or $x = \ln(-2 + \sqrt{5})$.

11 b)

$$\begin{aligned} \sinh^2 x - 2\cosh x - 7 &= 0 \\ (\cosh^2 x - 1) - 2\cosh x - 7 &= 0 \\ \cosh^2 x - 2\cosh x - 8 &= 0 \\ (\cosh x - 4)(\cosh x + 2) &= 0 \\ \cosh x = 4 \text{ or } \cosh x &= -2. \end{aligned}$$

$\cosh x = -2$ has no real solutions. If $\cosh x = 4$, then $x = \pm \ln(4 + \sqrt{4^2 - 1}) = \pm \ln(4 + \sqrt{15})$. So $x = \ln(4 + \sqrt{15})$ or $x = \ln(4 - \sqrt{15})$.

11 c)

$$\begin{aligned}
\cosh^2 x - 14\sinh x + 47 &= 0 \\
(\sinh^2 x + 1) - 14\sinh x + 47 &= 0 \\
\sinh^2 x - 14\sinh x + 48 &= 0 \\
(\sinh x - 6)(\sinh x - 8) &= 0 \\
\sinh x = 6 \text{ or } \sinh x &= 8.
\end{aligned}$$

If $\sinh x = 6$, then $x = \ln(6 + \sqrt{6^2 + 1}) = \ln(6 + \sqrt{37})$. If $\sinh x = 8$, then $x = \ln(8 + \sqrt{8^2 + 1}) = \ln(8 + \sqrt{65})$. So $x = \ln(6 + \sqrt{37})$ or $x = \ln(8 + \sqrt{65})$.

11 d)

$$\begin{aligned}
2\sinh^2 x - 13\cosh x + 8 &= 0 \\
2(\cosh^2 x - 1) - 13\cosh x + 8 &= 0 \\
2\cosh^2 x - 13\cosh x + 6 &= 0 \\
(2\cosh x - 1)(\cosh x - 6) &= 0 \\
\cosh x = \frac{1}{2} \text{ or } \cosh x &= 6.
\end{aligned}$$

If $\cosh x = \frac{1}{2}$, there are no real solutions as $\cosh x \geq 1$ for all $x \in \mathbb{R}$. If $\cosh x = 6$, then $x = \pm\ln(6 + \sqrt{6^2 - 1}) = \pm\ln(6 + \sqrt{35})$. So $x = \ln(6 + \sqrt{35})$ or $x = \ln(6 - \sqrt{35})$.

11 e)

$$\begin{aligned}
3\cosh^2 x + 22\sinh x - 19 &= 0 \\
3(\sinh^2 x + 1) + 22\sinh x - 19 &= 0 \\
3\sinh^2 x + 22\sinh x - 16 &= 0 \\
(3\sinh x - 2)(\sinh x + 8) &= 0 \\
\sinh x = \frac{2}{3} \text{ or } \sinh x &= -8.
\end{aligned}$$

If $\sinh x = \frac{2}{3}$, then $x = \ln\left(\frac{2}{3} + \sqrt{(\frac{2}{3})^2 + 1}\right) = \ln\left(\frac{2}{3} + \frac{\sqrt{13}}{3}\right)$. If $\sinh x = -8$, then $x = \ln(-8 + \sqrt{(-8)^2 + 1}) = \ln(-8 + \sqrt{65})$. So $x = \ln\left(\frac{2}{3} + \frac{\sqrt{13}}{3}\right)$ or $x = \ln(-8 + \sqrt{65})$.

11 f)

$$\begin{aligned}
4\sinh^2 x - 4\cosh x - 11 &= 0 \\
4(\cosh^2 x - 1) - 4\cosh x - 11 &= 0 \\
4\cosh^2 x - 4\cosh x - 15 &= 0 \\
(2\cosh x + 3)(2\cosh x - 5) &= 0 \\
\cosh x = -\frac{3}{2} \text{ or } \cosh x &= \frac{5}{2}.
\end{aligned}$$

If $\cosh x = -\frac{3}{2}$, there are no real solutions as $\cosh x \geq 1$ for all $x \in \mathbb{R}$. If $\cosh x = \frac{5}{2}$, then $x = \pm\ln\left(\frac{5}{2} + \sqrt{(\frac{5}{2})^2 - 1}\right) = \pm\ln\left(\frac{5}{2} + \frac{\sqrt{21}}{2}\right)$. So $x = \ln\left(\frac{5}{2} + \frac{\sqrt{21}}{2}\right)$ or $x = \ln\left(\frac{5}{2} - \frac{\sqrt{21}}{2}\right)$.

12 a)

$$\begin{aligned} 25\cosh^4 x - 40\cosh^2 x - 9 &= 0 \\ (5\cosh^2 x + 1)(5\cosh^2 x - 9) &= 0 \\ \cosh^2 x = -\frac{1}{5} \text{ or } \cosh^2 x &= \frac{9}{5}. \end{aligned}$$

$\cosh^2 x = -\frac{1}{5}$ has no real solutions as $\cosh^2 x \geq 1$. Also $\cosh^2 x = \frac{9}{5} \Rightarrow \cosh x = \pm\frac{3\sqrt{5}}{5}$. The negative value has no real solutions as $\cosh x \geq 1$. If $\cosh x = \frac{3\sqrt{5}}{5}$, then

$$x = \pm \ln\left(\frac{3\sqrt{5}}{5} + \sqrt{(\frac{3\sqrt{5}}{5})^2 - 1}\right) = \pm \ln\left(\frac{3\sqrt{5}}{5} + \frac{2\sqrt{5}}{5}\right) = \pm \ln\sqrt{5}.$$

So $x = \pm \ln\sqrt{5}$.

12 b)

$$\begin{aligned} 36\cosh^4 x - 36\sinh^2 x - 91 &= 0 \\ 36\cosh^4 x - 36(\cosh^2 x - 1) - 91 &= 0 \\ 36\cosh^4 x - 36\cosh^2 x - 55 &= 0 \\ (6\cosh^2 x + 5)(6\cosh^2 x - 11) &= 0 \\ \cosh^2 x = -\frac{5}{6} \text{ or } \cosh^2 x &= \frac{11}{6}. \end{aligned}$$

$\cosh^2 x = -\frac{5}{6}$ has no real solutions as $\cosh^2 x \geq 1$. From $\cosh^2 x = \frac{11}{6}$, $\cosh x = \pm\frac{\sqrt{66}}{6}$; the negative value has no real solutions as $\cosh x \geq 1$. If $\cosh x = \frac{\sqrt{66}}{6}$, then

$$x = \pm \ln\left(\frac{\sqrt{66}}{6} + \sqrt{(\frac{\sqrt{66}}{6})^2 - 1}\right) = \pm \ln\left(\frac{\sqrt{66} + \sqrt{30}}{6}\right).$$

So $x = \pm \ln\left(\frac{\sqrt{66}+\sqrt{30}}{6}\right)$.

12 c)

$$\begin{aligned} 192\sinh^4 x - 172\cosh^2 x + 67 &= 0 \\ 192\sinh^4 x - 172(1 + \sinh^2 x) + 67 &= 0 \\ 192\sinh^4 x - 172\sinh^2 x - 105 &= 0 \\ (16\sinh^2 x - 21)(12\sinh^2 x + 5) &= 0 \\ \sinh^2 x = \frac{21}{16} \text{ or } \sinh^2 x &= -\frac{5}{12}. \end{aligned}$$

$\sinh^2 x = -\frac{5}{12}$ has no real solutions as $\sinh^2 x \geq 0$. From $\sinh^2 x = \frac{21}{16}$, $\sinh x = \pm\frac{\sqrt{21}}{4}$.

$$x = \ln\left(\frac{\sqrt{21}}{4} + \sqrt{(\frac{\sqrt{21}}{4})^2 + 1}\right) = \ln\left(\frac{\sqrt{21} + \sqrt{37}}{4}\right).$$

For $\sinh x = -\frac{\sqrt{21}}{4}$,

$$x = \ln\left(-\frac{\sqrt{21}}{4} + \sqrt{(\frac{\sqrt{21}}{4})^2 + 1}\right) = \ln\left(\frac{\sqrt{37} - \sqrt{21}}{4}\right).$$

So $x = \ln\left(\frac{\sqrt{21}+\sqrt{37}}{4}\right)$ or $x = \ln\left(\frac{\sqrt{37}-\sqrt{21}}{4}\right)$.

13 a)

$$\begin{aligned}\cosh^2 x - \sinh^2 x &\equiv 1\\ \frac{\cosh^2 x}{\sinh^2 x} - \frac{\sinh^2 x}{\sinh^2 x} &\equiv \frac{1}{\sinh^2 x}\\ \coth^2 x - 1 &\equiv \operatorname{cosech}^2 x.\end{aligned}$$

13 b)

$$\begin{aligned}\cosh^2 x - \sinh^2 x &\equiv 1\\ \frac{\cosh^2 x}{\cosh^2 x} - \frac{\sinh^2 x}{\cosh^2 x} &\equiv \frac{1}{\cosh^2 x}\\ 1 - \tanh^2 x &\equiv \operatorname{sech}^2 x.\end{aligned}$$

14 a)

$$\begin{aligned}2\operatorname{sech}^2 x - 3\tanh x &= 0\\ 2(1 - \tanh^2 x) - 3\tanh x &= 0\\ 2\tanh^2 x + 3\tanh x - 2 &= 0\\ (2\tanh x - 1)(\tanh x + 2) &= 0\\ \tanh x = \frac{1}{2} \text{ or } \tanh x &= -2.\end{aligned}$$

If $\tanh x = \frac{1}{2}$, then $x = \frac{1}{2}\ln\left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\right) = \frac{1}{2}\ln 3$. If $\tanh x = -2$, there are no real solutions as $-1 < \tanh x < 1$ for all $x \in \mathbb{R}$. So $x = \frac{1}{2}\ln 3$.

14 b)

$$\begin{aligned}3\tanh^2 x - 5\operatorname{sech} x - 1 &= 0\\ 3(1 - \operatorname{sech}^2 x) - 5\operatorname{sech} x - 1 &= 0\\ 3\operatorname{sech}^2 x + 5\operatorname{sech} x - 2 &= 0\\ (3\operatorname{sech} x - 1)(\operatorname{sech} x + 2) &= 0\\ \operatorname{sech} x = \frac{1}{3} \text{ or } \operatorname{sech} x &= -2\\ \cosh x = 3 \text{ or } \cosh x &= -\frac{1}{2}.\end{aligned}$$

If $\cosh x = 3$, then $x = \pm\ln(3 + \sqrt{3^2 - 1}) = \pm\ln(3 + 2\sqrt{2})$. If $\cosh x = -\frac{1}{2}$, there are no real solutions as $\cosh x \geq 1$ for all $x \in \mathbb{R}$. So $x = \ln(3 + 2\sqrt{2})$ or $x = \ln(3 - 2\sqrt{2})$.

14 c)

$$\begin{aligned}
\operatorname{cosech}^2 x - 2\coth x - 7 &= 0 \\
(\coth^2 x - 1) - 2\coth x - 7 &= 0 \\
\coth^2 x - 2\coth x - 8 &= 0 \\
(\coth x - 4)(\coth x + 2) &= 0 \\
\coth x = 4 \text{ or } \coth x &= -2 \\
\tanh x = \frac{1}{4} \text{ or } \tanh x &= -\frac{1}{2}.
\end{aligned}$$

If $\tanh x = \frac{1}{4}$, then $x = \frac{1}{2}\ln\left(\frac{1+\frac{1}{4}}{1-\frac{1}{4}}\right) = \frac{1}{2}\ln\frac{5}{3}$. If $\tanh x = -\frac{1}{2}$, then $x = \frac{1}{2}\ln\left(\frac{1+(-\frac{1}{2})}{1-(-\frac{1}{2})}\right) = \frac{1}{2}\ln\frac{1}{3}$. So $x = \frac{1}{2}\ln\frac{5}{3}$ or $x = \frac{1}{2}\ln\frac{1}{3}$.

14 d)

$$\begin{aligned}
4\coth^2 x - 25\operatorname{cosech} x + 2 &= 0 \\
4(\operatorname{cosech}^2 x + 1) - 25\operatorname{cosech} x + 2 &= 0 \\
4\operatorname{cosech}^2 x - 25\operatorname{cosech} x + 6 &= 0 \\
(4\operatorname{cosech} x - 1)(\operatorname{cosech} x - 6) &= 0 \\
\operatorname{cosech} x = \frac{1}{4} \text{ or } \operatorname{cosech} x &= 6 \\
\sinh x = 4 \text{ or } \sinh x &= \frac{1}{6}.
\end{aligned}$$

If $\sinh x = 4$, then $x = \ln(4+\sqrt{4^2+1}) = \ln(4+\sqrt{17})$. If $\sinh x = \frac{1}{6}$, then $x = \ln\left(\frac{1}{6} + \sqrt{(\frac{1}{6})^2+1}\right) = \ln\left(\frac{1}{6} + \frac{\sqrt{37}}{6}\right)$. So $x = \ln(4+\sqrt{17})$ or $x = \ln\left(\frac{1}{6} + \frac{\sqrt{37}}{6}\right)$.

15 a)

$$\begin{aligned}
8\operatorname{sech}^2 x - 6\tanh x + 1 &= 0 \\
8(1 - \tanh^2 x) - 6\tanh x + 1 &= 0 \\
8\tanh^2 x + 6\tanh x - 9 &= 0 \\
(4\tanh x - 3)(2\tanh x + 3) &= 0 \\
\tanh x = \frac{3}{4} \text{ or } \tanh x &= -\frac{3}{2}.
\end{aligned}$$

If $\tanh x = \frac{3}{4}$, then $x = \frac{1}{2}\ln\left(\frac{1+\frac{3}{4}}{1-\frac{3}{4}}\right) = \frac{1}{2}\ln 7$. $\tanh x = -\frac{3}{2}$ has no real solutions as $-1 < \tanh x < 1$. So $x = \frac{1}{2}\ln 7$.

15 b)

$$\begin{aligned}
10\tanh^2 x - 21\operatorname{sech} x + 3 &= 0 \\
10(1 - \operatorname{sech}^2 x) - 21\operatorname{sech} x + 3 &= 0
\end{aligned}$$

$$
\begin{aligned}
10\operatorname{sech}^2 x + 21\operatorname{sech} x - 13 &= 0 \\
(5\operatorname{sech} x + 13)(2\operatorname{sech} x - 1) &= 0 \\
\operatorname{sech} x = -\frac{13}{5} \text{ or } \operatorname{sech} x &= \frac{1}{2} \\
\cosh x = -\frac{5}{13} \text{ or } \cosh x &= 2.
\end{aligned}
$$

$\cosh x = -\frac{5}{13}$ has no real solutions as $\cosh x \geq 1$. If $\cosh x = 2$, then $x = \pm\ln(2 + \sqrt{2^2 - 1}) = \pm\ln(2 + \sqrt{3})$. So $x = \ln(2 + \sqrt{3})$ or $x = \ln(2 - \sqrt{3})$.

15 c)

$$
\begin{aligned}
24\operatorname{cosech}^2 x + 58\coth x + 59 &= 0 \\
24(\coth^2 x - 1) + 58\coth x + 59 &= 0 \\
24\coth^2 x + 58\coth x + 35 &= 0 \\
(6\coth x + 7)(4\coth x + 5) &= 0 \\
\coth x = -\frac{7}{6} \text{ or } \coth x &= -\frac{5}{4}.
\end{aligned}
$$

If $\coth x = -\frac{7}{6}$, then $\tanh x = -\frac{6}{7}$ and $x = \frac{1}{2}\ln\left(\frac{1+(-\frac{6}{7})}{1-\frac{6}{7}}\right) = \frac{1}{2}\ln\frac{1}{13}$. If $\coth x = -\frac{5}{4}$, then $\tanh x = -\frac{4}{5}$ and $x = \frac{1}{2}\ln\left(\frac{1+(-\frac{4}{5})}{1-(-\frac{4}{5})}\right) = \frac{1}{2}\ln\frac{1}{9}$. So $x = \frac{1}{2}\ln\frac{1}{13}$ or $x = \frac{1}{2}\ln\frac{1}{9}$.

15 d)

$$
\begin{aligned}
25\coth^2 x - 73\operatorname{cosech} x &= 31 \\
25(\operatorname{cosech}^2 x + 1) - 73\operatorname{cosech} x &= 31 \\
25\operatorname{cosech}^2 x - 73\operatorname{cosech} x - 6 &= 0 \\
(25\operatorname{cosech} x + 2)(\operatorname{cosech} x - 3) &= 0 \\
\operatorname{cosech} x = -\frac{2}{25} \text{ or } \operatorname{cosech} x &= 3.
\end{aligned}
$$

If $\operatorname{cosech} x = -\frac{2}{25}$, then $\sinh x = -\frac{25}{2}$ and $x = \ln\left(-\frac{25}{2} + \sqrt{(-\frac{25}{2})^2 + 1}\right) = \ln\left(-\frac{25}{2} + \frac{\sqrt{629}}{2}\right)$. If $\operatorname{cosech} x = 3$, then $\sinh x = \frac{1}{3}$ and $x = \ln\left(\frac{1}{3} + \sqrt{(\frac{1}{3})^2 + 1}\right) = \ln\left(\frac{1}{3} + \frac{\sqrt{10}}{3}\right)$. So $x = \ln\left(-\frac{25}{2} + \frac{\sqrt{629}}{2}\right)$ or $x = \ln\left(\frac{1}{3} + \frac{\sqrt{10}}{3}\right)$.

16)

$$
\begin{aligned}
6\tanh^2 x + 17\operatorname{sech} x - 18 &= 0 \\
6(1 - \operatorname{sech}^2 x) + 17\operatorname{sech} x - 18 &= 0 \\
6\operatorname{sech}^2 x - 17\operatorname{sech} x + 12 &= 0 \\
(3\operatorname{sech} x - 4)(2\operatorname{sech} x - 3) &= 0 \\
\operatorname{sech} x = \frac{4}{3} \text{ or } \operatorname{sech} x &= \frac{3}{2}.
\end{aligned}
$$

However, $0 < \operatorname{sech} x \leq 1$, so $\operatorname{sech} x = \frac{4}{3}$ and $\operatorname{sech} x = \frac{3}{2}$ have no real solutions.

17 a)

$$\begin{aligned}
\sinh x &= \operatorname{sech} x \\
\sinh x &= \frac{1}{\cosh x} \\
\sinh x \cosh x &= 1 \\
\frac{1}{2}\sinh 2x &= 1 \\
\sinh 2x &= 2 \\
2x &= \ln(2+\sqrt{2^2+1}) \\
x &= \frac{1}{2}\ln(2+\sqrt{5}).
\end{aligned}$$

17 b)

$$\begin{aligned}
\tanh x &= \operatorname{sech} x \\
\frac{\sinh x}{\cosh x} &= \frac{1}{\cosh x} \\
\sinh x &= 1 \\
x &= \ln(1+\sqrt{1^2+1}) = \ln(1+\sqrt{2}).
\end{aligned}$$

17 c)

$$\begin{aligned}
\sinh x &= 2\coth x \\
\sinh x &= 2\frac{\cosh x}{\sinh x} \\
\sinh^2 x &= 2\cosh x \\
\sinh^2 x - 2\cosh x &= 0 \\
(\cosh^2 x - 1) - 2\cosh x &= 0 \\
\cosh^2 x - 2\cosh x - 1 &= 0 \\
(\cosh x - 1)^2 - 2 &= 0 \\
(\cosh x - 1)^2 &= 2 \\
\cosh x - 1 &= \pm\sqrt{2} \\
\cosh x &= 1 \pm \sqrt{2}.
\end{aligned}$$

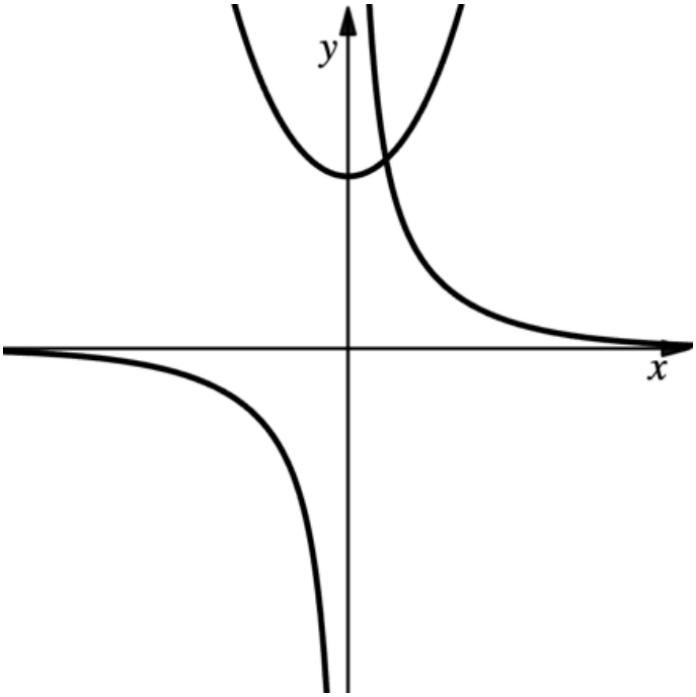
If $\cosh x = 1+\sqrt{2}$, then $x = \pm\ln\left(1+\sqrt{2}+\sqrt{(1+\sqrt{2})^2-1}\right) = \pm\ln\left(1+\sqrt{2}+\sqrt{2+2\sqrt{2}}\right)$. If $\cosh x = 1-\sqrt{2}$, there are no real solutions as $\cosh x \geq 1$ for all $x \in \mathbb{R}$. So $x = \ln(1+\sqrt{2}+\sqrt{2+2\sqrt{2}})$ or $x = \ln(1+\sqrt{2}-\sqrt{2+2\sqrt{2}})$.

17 d)

$$\begin{aligned} 3\cosh x &= \coth x \\ 3\cosh x &= \frac{\cosh x}{\sinh x} \\ 3\sinh x\cosh x - \cosh x &= 0 \\ \cosh x(3\sinh x - 1) &= 0 \\ \cosh x = 0 \text{ or } \sinh x &= \frac{1}{3}. \end{aligned}$$

If $\cosh x = 0$, there are no real solutions as $\cosh x \geq 1$ for all $x \in \mathbb{R}$. If $\sinh x = \frac{1}{3}$, then $x = \ln\left(\frac{1}{3} + \sqrt{(\frac{1}{3})^2 + 1}\right) = \ln\left(\frac{1}{3} + \frac{\sqrt{10}}{3}\right)$.

18 a)



18 b)

$$\begin{aligned} 2\cosh x &= \operatorname{cosech} x \\ 2\cosh x &= \frac{1}{\sinh x} \\ 2\sinh x\cosh x &= 1 \\ \sinh 2x &= 1 \\ 2x &= \ln(1 + \sqrt{1^2 + 1}) \\ x &= \frac{1}{2}\ln(1 + \sqrt{2}). \end{aligned}$$

Let $y = 2\cosh\left(\frac{1}{2}\ln(1+\sqrt{2})\right)$.

$$y = 2 \times \frac{e^{\frac{1}{2}\ln(1+\sqrt{2})} + e^{-\frac{1}{2}\ln(1+\sqrt{2})}}{2}$$

$$
\begin{aligned}
&= e^{\frac{1}{2}\ln(1+\sqrt{2})} + e^{-\frac{1}{2}\ln(1+\sqrt{2})} \\
&= e^{\ln((1+\sqrt{2})^{1/2})} + e^{\ln((1+\sqrt{2})^{-1/2})} \\
&= (1+\sqrt{2})^{1/2} + (1+\sqrt{2})^{-1/2} \\
&= \sqrt{1+\sqrt{2}} + \frac{1}{\sqrt{1+\sqrt{2}}} \\
&= \frac{1+\sqrt{2}}{\sqrt{1+\sqrt{2}}} + \frac{1}{\sqrt{1+\sqrt{2}}} \\
&= \frac{2+\sqrt{2}}{\sqrt{1+\sqrt{2}}}.
\end{aligned}
$$

So the two graphs intersect at $\left(\frac{1}{2}\ln(1+\sqrt{2}), \frac{2+\sqrt{2}}{\sqrt{1+\sqrt{2}}}\right)$. **Alternatively:**

$$
\begin{aligned}
2\cosh x &= \operatorname{cosech} x \\
2\left(\frac{e^x + e^{-x}}{2}\right) &= \frac{2}{e^x - e^{-x}} \\
(e^x + e^{-x})(e^x - e^{-x}) &= 2 \\
e^{2x} - e^{-2x} &= 2 \\
e^{2x} - e^{-2x} - 2 &= 0 \\
e^{4x} - 2e^{2x} - 1 &= 0 \\
(e^{2x} - 1)^2 - 2 &= 0 \\
(e^{2x} - 1)^2 &= 2 \\
e^{2x} - 1 &= \pm\sqrt{2} \\
e^{2x} &= 1 \pm \sqrt{2} \\
2x &= \ln(1 \pm \sqrt{2}) \\
x &= \frac{1}{2}\ln(1 \pm \sqrt{2}).
\end{aligned}
$$

Intersect at positive value: $x = \frac{1}{2}\ln(1+\sqrt{2})$. Then

$$
\begin{aligned}
y = 2\cosh\left(\frac{1}{2}\ln(1+\sqrt{2})\right) &= 2\cosh(\ln\sqrt{1+\sqrt{2}}) \\
&= 2 \times \frac{e^{\ln\sqrt{1+\sqrt{2}}} + e^{-\ln\sqrt{1+\sqrt{2}}}}{2} \\
&= \sqrt{1+\sqrt{2}} + \frac{1}{\sqrt{1+\sqrt{2}}} \\
&= \frac{2+\sqrt{2}}{\sqrt{1+\sqrt{2}}} \\
&= \frac{(2+\sqrt{2})\sqrt{1+\sqrt{2}}}{1+\sqrt{2}} \times \frac{1-\sqrt{2}}{1-\sqrt{2}} \\
&= \sqrt{2}\sqrt{1+\sqrt{2}} = \sqrt{2(1+\sqrt{2})}.
\end{aligned}
$$

So the two graphs intersect at $\left(\frac{1}{2}\ln(1+\sqrt{2}), \sqrt{2(1+\sqrt{2})}\right)$.

19 a)

$$\begin{aligned}
\sinh x &= 3 \\
\cosh^2 x - \sinh^2 x &= 1 \\
\cosh^2 x - 3^2 &= 1 \\
\cosh^2 x &= 10 \\
\cosh x &= \sqrt{10}.
\end{aligned}$$

19 b)

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{3}{\sqrt{10}} = \frac{3\sqrt{10}}{10}.$$

19 c)

$$\coth x = \frac{\cosh x}{\sinh x} = \frac{\sqrt{10}}{3}.$$

19 d)

$$\operatorname{cosech} x = \frac{1}{\sinh x} = \frac{1}{3}.$$

19 e)

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{1}{\sqrt{10}} = \frac{\sqrt{10}}{10}.$$

20 a)

$$\begin{aligned}
1 - \tanh^2 x &= \operatorname{sech}^2 x \\
1 - \tanh^2 x &= \left(\frac{1}{\sqrt{2}}\right)^2 \\
\tanh^2 x &= \frac{1}{2} \\
\tanh x &= \pm\frac{1}{\sqrt{2}}.
\end{aligned}$$

20 b)

$$\cosh x = \frac{1}{\operatorname{sech} x} = \frac{1}{1/\sqrt{2}} = \sqrt{2}.$$

20 c)

$$\begin{aligned}\cosh^2 x - \sinh^2 x &= 1\\ (\sqrt{2})^2 - \sinh^2 x &= 1\\ \sinh^2 x &= 1\\ \sinh x &= \pm 1.\end{aligned}$$

20 d)

$$\operatorname{cosech} x = \frac{1}{\sinh x} = \frac{1}{\pm 1} = \pm 1.$$

20 e)

$$\coth x = \frac{1}{\tanh x} = \frac{1}{\pm 1/\sqrt{2}} = \pm\sqrt{2}.$$

21 a)

$$\begin{aligned}48\coth^2 x + 34\operatorname{cosech} x - 53 &= 0\\ 48(\operatorname{cosech}^2 x + 1) + 34\operatorname{cosech} x - 53 &= 0\\ 48\operatorname{cosech}^2 x + 34\operatorname{cosech} x - 5 &= 0\\ (6\operatorname{cosech} x + 5)(8\operatorname{cosech} x - 1) &= 0\\ \operatorname{cosech} x = -\frac{5}{6} \text{ or } \operatorname{cosech} x &= \frac{1}{8}.\end{aligned}$$

But $\operatorname{cosech} x = \frac{1}{8}$ only as $x > 0$.

21 b)

$$\sinh x = \frac{1}{\operatorname{cosech} x} = \frac{1}{1/8} = 8.$$

21 c)

$$\begin{aligned}\cosh^2 x - \sinh^2 x &= 1\\ \cosh^2 x - 8^2 &= 1\\ \cosh^2 x &= 65\\ \cosh x &= \pm\sqrt{65}.\end{aligned}$$

But $\cosh x = \sqrt{65}$ as $\cosh x \geq 1$ for all real values of x.

21 d)

$$\operatorname{sech} x = \frac{1}{\cosh x} = \frac{1}{\sqrt{65}} = \frac{\sqrt{65}}{65}.$$

21 e)

$$\tanh x = \frac{\sinh x}{\cosh x} = \frac{8}{\sqrt{65}} = \frac{8\sqrt{65}}{65}.$$

21 f)

$$\coth x = \frac{1}{\tanh x} = \frac{65}{8\sqrt{65}} = \frac{\sqrt{65}}{8}.$$

22 a)

$$\frac{\operatorname{sech}^2 x}{1 - \tanh^2 x} \equiv \frac{\operatorname{sech}^2 x}{\operatorname{sech}^2 x} \equiv 1.$$

22 b)

$$\frac{\coth^2 x - 1}{\operatorname{cosech}^2 x} \equiv \frac{\operatorname{cosech}^2 x}{\operatorname{cosech}^2 x} \equiv 1.$$

22 c)

$$\begin{aligned}
\frac{1 - \tanh^2 x}{\coth^2 x - 1} &\equiv \frac{\operatorname{sech}^2 x}{\operatorname{cosech}^2 x} \\
&\equiv \frac{1/\cosh^2 x}{1/\sinh^2 x} \\
&\equiv \frac{1}{\cosh^2 x} \times \frac{\sinh^2 x}{1} \\
&\equiv \tanh^2 x.
\end{aligned}$$

22 d)

$$\begin{aligned}
\frac{3 - 3\coth^2 x}{8 - 8\tanh^2 x} &\equiv -\frac{3(\coth^2 x - 1)}{8(1 - \tanh^2 x)} \\
&\equiv -\frac{3\operatorname{cosech}^2 x}{8\operatorname{sech}^2 x} \\
&\equiv -\frac{3/\sinh^2 x}{8/\cosh^2 x} \\
&\equiv -\frac{3}{\sinh^2 x} \times \frac{\cosh^2 x}{8} \\
&\equiv -\frac{3}{8}\coth^2 x.
\end{aligned}$$

23 a)

$$\begin{aligned}
\cosh^2 x - \sinh^2 x &= 1 \\
\cosh^2(1+\ln 6) - \sinh^2(1+\ln 6) &= 1 \\
\cosh^2(1+\ln 6) - \left(\frac{36e^2-1}{12e}\right)^2 &= 1 \\
\cosh^2(1+\ln 6) &= 1 + \frac{(36e^2-1)^2}{(12e)^2} \\
&= \frac{(12e)^2 + (36e^2-1)^2}{(12e)^2} \\
&= \frac{144e^2 + 1296e^4 - 72e^2 + 1}{(12e)^2} \\
&= \frac{1296e^4 + 72e^2 + 1}{(12e)^2} \\
&= \frac{(36e^2+1)^2}{(12e)^2} \\
\cosh(1+\ln 6) &= \frac{36e^2+1}{12e}.
\end{aligned}$$

23 b)

$$\begin{aligned}
\tanh(1+\ln 6) &= \frac{\sinh(1+\ln 6)}{\cosh(1+\ln 6)} \\
&= \frac{\frac{36e^2-1}{12e}}{\frac{36e^2+1}{12e}} \\
&= \frac{36e^2-1}{36e^2+1}.
\end{aligned}$$

24 a)

$$\begin{aligned}
p\left(\frac{5}{4}\right) &= 12\left(\frac{5}{4}\right)^3 - 7\left(\frac{5}{4}\right)^2 - 42\left(\frac{5}{4}\right) + 40 \\
&= \frac{375}{16} - \frac{175}{16} - \frac{105}{2} + 40 \\
&= \frac{375}{16} - \frac{175}{16} - \frac{840}{16} + \frac{640}{16} \\
&= \frac{200}{16} - \frac{840}{16} + \frac{640}{16} \\
&= -\frac{640}{16} + \frac{640}{16} = 0.
\end{aligned}$$

Therefore, $(4x-5)$ is a factor of $p(x)$.

24 b)

$$\begin{aligned} 12x^3 - 7x^2 - 42x + 40 &= (4x-5)(3x^2+2x-8) \\ &= (4x-5)(3x-4)(x+2). \end{aligned}$$

24 c)

$$\begin{aligned} 12\operatorname{sech}^3\theta + 7\tanh^2\theta - 42\operatorname{sech}\theta + 33 &= 0 \\ 12\operatorname{sech}^3\theta + 7(1-\operatorname{sech}^2\theta) - 42\operatorname{sech}\theta + 33 &= 0 \\ 12\operatorname{sech}^3\theta - 7\operatorname{sech}^2\theta - 42\operatorname{sech}\theta + 40 &= 0 \\ (4\operatorname{sech}\theta - 5)(3\operatorname{sech}\theta - 4)(\operatorname{sech}\theta + 2) &= 0 \\ \operatorname{sech}\theta = \frac{5}{4} \text{ or } \operatorname{sech}\theta = \frac{4}{3} \text{ or } \operatorname{sech}\theta &= -2. \end{aligned}$$

However, $0 < \operatorname{sech}\theta \leq 1$, so $\operatorname{sech}\theta = \frac{5}{4}$, $\operatorname{sech}\theta = \frac{4}{3}$ and $\operatorname{sech}\theta = -2$ have no real solutions.

25 a)

$$\begin{aligned} \sinh(\ln 3) &= \frac{e^{\ln 3} - e^{-\ln 3}}{2} = \frac{e^{\ln 3} - e^{\ln(1/3)}}{2} = \frac{3 - 1/3}{2} = \frac{4}{3}, \\ \cosh(\ln 3) &= \frac{e^{\ln 3} + e^{-\ln 3}}{2} = \frac{e^{\ln 3} + e^{\ln(1/3)}}{2} = \frac{3 + 1/3}{2} = \frac{5}{3}. \end{aligned}$$

25 b)

$$\begin{aligned} \sinh(\ln 4) &= \frac{e^{\ln 4} - e^{-\ln 4}}{2} = \frac{e^{\ln 4} - e^{\ln(1/4)}}{2} = \frac{4 - 1/4}{2} = \frac{15}{8}, \\ \cosh(\ln 4) &= \frac{e^{\ln 4} + e^{-\ln 4}}{2} = \frac{e^{\ln 4} + e^{\ln(1/4)}}{2} = \frac{4 + 1/4}{2} = \frac{17}{8}. \end{aligned}$$

25 c)

$$\begin{aligned} \sinh(\ln 4 - \ln 3) &= \sinh(\ln 4)\cosh(\ln 3) - \cosh(\ln 4)\sinh(\ln 3) \\ &= \frac{15}{8} \times \frac{5}{3} - \frac{17}{8} \times \frac{4}{3} = \frac{25}{8} - \frac{17}{6} = \frac{7}{24}. \end{aligned}$$

26 a)

$$\begin{aligned} \sinh(\ln 5) &= \frac{e^{\ln 5} - e^{-\ln 5}}{2} = \frac{e^{\ln 5} - e^{\ln(1/5)}}{2} = \frac{5 - 1/5}{2} = \frac{12}{5}, \\ \cosh(\ln 5) &= \frac{e^{\ln 5} + e^{-\ln 5}}{2} = \frac{e^{\ln 5} + e^{\ln(1/5)}}{2} = \frac{5 + 1/5}{2} = \frac{13}{5}. \end{aligned}$$

26 b)

$$\sinh(\ln 6) = \frac{e^{\ln 6} - e^{-\ln 6}}{2} = \frac{e^{\ln 6} - e^{\ln(1/6)}}{2} = \frac{6 - 1/6}{2} = \frac{35}{12},$$
$$\cosh(\ln 6) = \frac{e^{\ln 6} + e^{-\ln 6}}{2} = \frac{e^{\ln 6} + e^{\ln(1/6)}}{2} = \frac{6 + 1/6}{2} = \frac{37}{12}.$$

26 c)

$$\begin{aligned}\cosh(\ln 6 - \ln 5) &= \cosh(\ln 6)\cosh(\ln 5) - \sinh(\ln 6)\sinh(\ln 5) \\ &= \frac{37}{12} \times \frac{13}{5} - \frac{35}{12} \times \frac{12}{5} = \frac{481}{60} - 7 = \frac{61}{60}.\end{aligned}$$

27 a)

$$\tanh(\ln 2) = \frac{e^{\ln 2} - e^{-\ln 2}}{e^{\ln 2} + e^{-\ln 2}} = \frac{2 - \frac{1}{2}}{2 + \frac{1}{2}} = \frac{3}{5}.$$

27 b)

$$\tanh(\ln 7) = \frac{e^{\ln 7} - e^{-\ln 7}}{e^{\ln 7} + e^{-\ln 7}} = \frac{7 - \frac{1}{7}}{7 + \frac{1}{7}} = \frac{24}{25}.$$

27 c)

$$\begin{aligned}\tanh(\ln 2 + \ln 7) &= \frac{\tanh(\ln 2) + \tanh(\ln 7)}{1 + \tanh(\ln 2)\tanh(\ln 7)} \\ &= \frac{\frac{3}{5} + \frac{24}{25}}{1 + \frac{3}{5} \times \frac{24}{25}} = \frac{195}{197}.\end{aligned}$$

28 a)

$$\sinh 3 \cosh 6 - \cosh 3 \sinh 6 = \sinh(3 - 6) = \sinh(-3).$$

28 b)

$$\sinh 4 \cosh 5 + \cos 4 \sinh 5 = \sinh(4 + 5) = \sinh 9.$$

29 a)

$$\cosh 3 \cosh 7 - \sinh 3 \sinh 7 = \cosh(3-7) = \cosh(-4).$$

29 b)

$$\cosh 8 \cosh 9 + \sinh 8 \sinh 9 = \cosh(8+9) = \cosh 17.$$

30 a)

$$\cosh x = 2 \Rightarrow 2^2 - \sinh^2 x = 1 \Rightarrow \sinh^2 x = 3 \Rightarrow \sinh x = \sqrt{3},$$
$$\cosh y = 3 \Rightarrow 3^2 - \sinh^2 y = 1 \Rightarrow \sinh^2 y = 8 \Rightarrow \sinh y = 2\sqrt{2}.$$

We must have $\sinh x > 0$ as $\cosh x = 2 > 0$. We must have $\sinh y > 0$ as $\cosh y = 3 > 0$. Therefore,

$$\cosh(x-y) = \cosh x \cosh y - \sinh x \sinh y = 2 \times 3 - \sqrt{3} \times 2\sqrt{2} = 6 - 2\sqrt{6}.$$

30 b)

$$\sinh(x-y) = \sinh x \cosh y - \sinh y \cosh x = \sqrt{3} \times 3 - 2\sqrt{2} \times 2 = 3\sqrt{3} - 4\sqrt{2}.$$

31 a)

$$\begin{aligned}\sinh(x+\ln 2) + \cosh(x-\ln 2) &= (\sinh x \cosh(\ln 2) + \cosh x \sinh(\ln 2)) \\ &\quad + (\cosh x \cosh(\ln 2) - \sinh x \sinh(\ln 2)) \\ &= \frac{5}{4}\sinh x + \frac{3}{4}\cosh x + \frac{5}{4}\cosh x - \frac{3}{4}\sinh x \\ &= \frac{1}{2}\sinh x + 2\cosh x.\end{aligned}$$

31 b)

$$\begin{aligned}\sinh(x-\ln 3) - \cosh(x+\ln 3) &= (\sinh x \cosh(\ln 3) - \cosh x \sinh(\ln 3)) \\ &\quad - (\cosh x \cosh(\ln 3) + \sinh x \sinh(\ln 3)) \\ &= \frac{5}{3}\sinh x - \frac{4}{3}\cosh x - \frac{5}{3}\cosh x - \frac{4}{3}\sinh x \\ &= \frac{1}{3}\sinh x - 3\cosh x.\end{aligned}$$

31 c)

$$\begin{aligned}\cosh(x-\ln 4) - \sinh(x-\ln 4) &= \cosh x \cosh(\ln 4) - \sinh x \sinh(\ln 4) \\ &\quad - (\sinh x \cosh(\ln 4) - \cosh x \sinh(\ln 4)) \\ &= \frac{17}{8}\cosh x - \frac{15}{8}\sinh x - \frac{17}{8}\sinh x + \frac{15}{8}\cosh x \\ &= 4\cosh x - 4\sinh x.\end{aligned}$$

32 a)

$$\begin{aligned}
\sinh(x+\ln 2) &= \cosh x \\
\sinh x\cosh(\ln 2)+\cosh x\sinh(\ln 2) &= \cosh x \\
\frac{5}{4}\sinh x+\frac{3}{4}\cosh x &= \cosh x \\
\frac{5}{4}\sinh x &= \frac{1}{4}\cosh x \\
5\sinh x &= \cosh x \\
\tanh x &= \frac{1}{5} \\
x &= \frac{1}{2}\ln\left(\frac{1+\frac{1}{5}}{1-\frac{1}{5}}\right)=\frac{1}{2}\ln\frac{3}{2}.
\end{aligned}$$

32 b)

$$\begin{aligned}
\cosh(x-\ln 3) &= \cosh x \\
\cosh x\cosh(\ln 3)-\sinh x\sinh(\ln 3) &= \cosh x \\
\frac{5}{3}\cosh x-\frac{4}{3}\sinh x &= \cosh x \\
\frac{2}{3}\cosh x &= \frac{4}{3}\sinh x \\
\cosh x &= 2\sinh x \\
\tanh x &= \frac{1}{2} \\
x &= \frac{1}{2}\ln\left(\frac{1+\frac{1}{2}}{1-\frac{1}{2}}\right)=\frac{1}{2}\ln 3.
\end{aligned}$$

32 c)

$$\begin{aligned}
\cosh(x-\ln 4) &= \sinh x \\
\cosh x\cosh(\ln 4)-\sinh x\sinh(\ln 4) &= \sinh x \\
\frac{17}{8}\cosh x-\frac{15}{8}\sinh x &= \sinh x \\
\frac{17}{8}\cosh x &= \frac{23}{8}\sinh x \\
17\cosh x &= 23\sinh x \\
\tanh x &= \frac{17}{23} \\
x &= \frac{1}{2}\ln\left(\frac{1+\frac{17}{23}}{1-\frac{17}{23}}\right)=\frac{1}{2}\ln\frac{20}{3}.
\end{aligned}$$

33 a)

$$\begin{aligned}
\sinh(x+\ln 2) &= \cosh(x-\ln 4)\\
\sinh x\cosh(\ln 2)+\cosh x\sinh(\ln 2) &= \cosh x\cosh(\ln 4)-\sinh x\sinh(\ln 4)\\
\frac{5}{4}\sinh x+\frac{3}{4}\cosh x &= \frac{17}{8}\cosh x-\frac{15}{8}\sinh x\\
\frac{25}{8}\sinh x &= \frac{11}{8}\cosh x\\
25\sinh x &= 11\cosh x\\
\tanh x &= \frac{11}{25}\\
x &= \frac{1}{2}\ln\left(\frac{1+\frac{11}{25}}{1-\frac{11}{25}}\right)=\frac{1}{2}\ln\frac{18}{7}.
\end{aligned}$$

33 b)

$$\begin{aligned}
\cosh(x-\ln 9) &= \sinh(x+\ln 3)\\
\cosh x\cosh(\ln 9)-\sinh x\sinh(\ln 9) &= \sinh x\cosh(\ln 3)+\cosh x\sinh(\ln 3)\\
\frac{41}{9}\cosh x-\frac{40}{9}\sinh x &= \frac{5}{3}\sinh x+\frac{4}{3}\cosh x\\
\frac{29}{9}\cosh x &= \frac{55}{9}\sinh x\\
29\cosh x &= 55\sinh x\\
\tanh x &= \frac{29}{55}\\
x &= \frac{1}{2}\ln\left(\frac{1+\frac{29}{55}}{1-\frac{29}{55}}\right)=\frac{1}{2}\ln\frac{42}{13}.
\end{aligned}$$

33 c)

$$\begin{aligned}
\cosh(x+3\ln 2) &= \cosh(x-\ln 20)\\
\cosh x\cosh(3\ln 2)+\sinh x\sinh(3\ln 2) &= \cosh x\cosh(\ln 20)-\sinh x\sinh(\ln 20)\\
\frac{65}{16}\cosh x+\frac{63}{16}\sinh x &= \frac{401}{40}\cosh x-\frac{399}{40}\sinh x\\
\frac{1113}{80}\sinh x &= \frac{477}{80}\cosh x\\
1113\sinh x &= 477\cosh x\\
\tanh x &= \frac{477}{1113}=\frac{3}{7}\\
x &= \frac{1}{2}\ln\left(\frac{1+\frac{3}{7}}{1-\frac{3}{7}}\right)=\frac{1}{2}\ln\frac{5}{2}.
\end{aligned}$$

34 a)

$$\begin{aligned}\text{LHS: } \coth x + \cosh x &\equiv \frac{\cosh x}{\sinh x} + \frac{\cosh x}{1} \\ &\equiv \frac{\cosh x + \sinh x \cosh x}{\sinh x} \\ &\equiv \frac{\cosh x(1 + \sinh x)}{\sinh x} \\ &\equiv \coth x(1 + \sinh x) \quad \text{: RHS.}\end{aligned}$$

34 b)

$$\begin{aligned}\text{LHS: } \coth x + \operatorname{sech} x &\equiv \frac{\cosh x}{\sinh x} + \frac{1}{\cosh x} \\ &\equiv \frac{\cosh^2 x + \sinh x}{\sinh x \cosh x} \\ &\equiv \frac{1}{\sinh x} \times \frac{1}{\cosh x}(\cosh^2 x + \sinh x) \\ &\equiv \operatorname{cosech} x \operatorname{sech} x(\sinh x + \cosh^2 x) \quad \text{: RHS.}\end{aligned}$$

34 c)

$$\begin{aligned}\text{LHS: } \coth x - \tanh x &\equiv \frac{\cosh x}{\sinh x} - \frac{\sinh x}{\cosh x} \\ &\equiv \frac{\cosh^2 x - \sinh^2 x}{\sinh x \cosh x} \\ &\equiv \frac{1}{\sinh x \cosh x} \\ &\equiv \operatorname{cosech} x \operatorname{sech} x \quad \text{: RHS.}\end{aligned}$$

35 a)

$$\begin{aligned}\text{LHS: } \cosh x - \operatorname{sech} x &\equiv \frac{\cosh x}{1} - \frac{1}{\cosh x} \\ &\equiv \frac{\cosh^2 x - 1}{\cosh x} \\ &\equiv \frac{\sinh^2 x}{\cosh x} \\ &\equiv \sinh x \tanh x \quad \text{: RHS.}\end{aligned}$$

35 b)

$$\text{LHS: } \operatorname{cosech} x + \sinh x \equiv \frac{1}{\sinh x} + \frac{\sinh x}{1}$$

$$\equiv \frac{1+\sinh^2 x}{\sinh x}$$
$$\equiv \frac{\cosh^2 x}{\sinh x}$$
$$\equiv \cosh x \coth x \quad : \text{RHS}.$$

35 c)

$$\text{RHS: } \tanh x(\coth^2 x - 1) \equiv \coth x - \tanh x$$
$$\equiv \frac{\cosh x}{\sinh x} - \frac{\sinh x}{\cosh x}$$
$$\equiv \frac{\cosh^2 x - \sinh^2 x}{\sinh x \cosh x}$$
$$\equiv \frac{1}{\sinh x \cosh x}$$
$$\equiv \operatorname{cosech} x \operatorname{sech} x \quad : \text{LHS}.$$

35 d)

$$\text{LHS: } \operatorname{cosech} x(\sinh x \tanh x - 1) \equiv \frac{1}{\sinh x}\left(\sinh x \times \frac{\sinh x}{\cosh x} - 1\right)$$
$$\equiv \frac{1}{\sinh x}\left(\frac{\sinh^2 x}{\cosh x} - 1\right)$$
$$\equiv \frac{\sinh x}{\cosh x} - \frac{1}{\sinh x}$$
$$\equiv \tanh x - \operatorname{cosech} x \quad : \text{RHS}.$$

36 a)

$$8 \sinh 2x \cosh 2x \equiv 4(2 \sinh 2x \cosh 2x) \equiv 4 \sinh(2 \times 2x) \equiv 4 \sinh 4x.$$

36 b)

$$2 \cosh^2 5x - 1 \equiv \cosh(2 \times 5x) \equiv \cosh 10x.$$

36 c)

$$1 + 2 \sinh^2 3x \equiv \cosh(2 \times 3x) \equiv \cosh 6x.$$

36 d)

$$9 \cosh^2 \frac{x}{4} + 9 \sinh^2 \frac{x}{4} \equiv 9\left(\cosh^2 \frac{x}{4} + \sinh^2 \frac{x}{4}\right) \equiv 9 \cosh\left(2 \times \frac{x}{4}\right) \equiv 9 \cosh \frac{x}{2}.$$

37 a)

$$\begin{aligned}\sinh 2\theta - 2\sinh\theta &= 0\\ 2\sinh\theta\cosh\theta - 2\sinh\theta &= 0\\ \sinh\theta(\cosh\theta - 1) &= 0\\ \sinh\theta = 0 \text{ or } \cosh\theta &= 1.\end{aligned}$$

If $\sinh\theta = 0$, then $\theta = \ln(0+\sqrt{0^2+1}) = \ln 1 = 0$. If $\cosh\theta = 1$, then $\theta = \ln(1+\sqrt{1^2-1}) = \ln 1 = 0$. So $\theta = 0$ is the only real solution.

37 b)

$$\begin{aligned}\sinh 2\theta &= 3\sinh\theta\\ 2\sinh\theta\cosh\theta - 3\sinh\theta &= 0\\ \sinh\theta(2\cosh\theta - 3) &= 0\\ \sinh\theta = 0 \text{ or } \cosh\theta &= \frac{3}{2}.\end{aligned}$$

If $\sinh\theta = 0$, then $\theta = 0$. If $\cosh\theta = \frac{3}{2}$, then $\theta = \pm\ln\left(\frac{3}{2} + \sqrt{(\frac{3}{2})^2 - 1}\right) = \pm\ln\left(\frac{3}{2} + \frac{\sqrt{5}}{2}\right)$. So $\theta = 0$, $\theta = \ln\left(\frac{3}{2} + \frac{\sqrt{5}}{2}\right)$ or $\theta = \ln\left(\frac{3}{2} - \frac{\sqrt{5}}{2}\right)$.

37 c)

$$\begin{aligned}5\sinh 2\theta &= 6\cosh\theta\\ 10\sinh\theta\cosh\theta - 6\cosh\theta &= 0\\ \cosh\theta(5\sinh\theta - 3) &= 0\\ \cosh\theta = 0 \text{ or } \sinh\theta &= \frac{3}{5}.\end{aligned}$$

$\cosh\theta = 0$ has no real solutions. If $\sinh\theta = \frac{3}{5}$, then $\theta = \ln\left(\frac{3}{5} + \sqrt{(\frac{3}{5})^2 + 1}\right) = \ln\left(\frac{3}{5} + \frac{\sqrt{34}}{5}\right)$. So $\theta = \ln\left(\frac{3}{5} + \frac{\sqrt{34}}{5}\right)$.

37 d)

$$\begin{aligned}8\cosh\theta - 3\sinh 2\theta &= 0\\ 8\cosh\theta - 6\sinh\theta\cosh\theta &= 0\\ \cosh\theta(4 - 3\sinh\theta) &= 0\\ \cosh\theta = 0 \text{ or } \sinh\theta &= \frac{4}{3}.\end{aligned}$$

$\cosh\theta = 0$ has no real solutions. If $\sinh\theta = \frac{4}{3}$, then $\theta = \ln\left(\frac{4}{3} + \sqrt{(\frac{4}{3})^2 + 1}\right) = \ln\left(\frac{4}{3} + \frac{5}{3}\right) = \ln 3$. So $\theta = \ln 3$.

38 a)

$$\begin{aligned} 2\cosh 2\theta &= 7\cosh\theta \\ 2\cosh 2\theta - 7\cosh\theta &= 0 \\ 2(2\cosh^2\theta - 1) - 7\cosh\theta &= 0 \\ 4\cosh^2\theta - 7\cosh\theta - 2 &= 0 \\ \cosh\theta = -\frac{1}{4} \text{ or } \cosh\theta &= 2. \end{aligned}$$

$\cosh\theta = -\frac{1}{4}$ has no real solutions. If $\cosh\theta = 2$, then $\theta = \pm\ln(2+\sqrt{2^2-1}) = \pm\ln(2+\sqrt{3})$. So $\theta = \ln(2+\sqrt{3})$ or $\theta = \ln(2-\sqrt{3})$.

38 b)

$$\begin{aligned} 3\cosh 2\theta + 11\sinh\theta &= 0 \\ 3(1+2\sinh^2\theta) + 11\sinh\theta &= 0 \\ 6\sinh^2\theta + 11\sinh\theta + 3 &= 0 \\ \sinh\theta = -\frac{3}{2} \text{ or } \sinh\theta &= -\frac{1}{3}. \end{aligned}$$

If $\sinh\theta = -\frac{3}{2}$, then $\theta = \ln\left(-\frac{3}{2}+\sqrt{(-\frac{3}{2})^2+1}\right) = \ln\left(-\frac{3}{2}+\frac{\sqrt{13}}{2}\right)$. If $\sinh\theta = -\frac{1}{3}$, then $\theta = \ln\left(-\frac{1}{3}+\sqrt{(-\frac{1}{3})^2+1}\right) = \ln\left(-\frac{1}{3}+\frac{\sqrt{10}}{3}\right)$. So $\theta = \ln\left(-\frac{3}{2}+\frac{\sqrt{13}}{2}\right)$ or $\theta = \ln\left(-\frac{1}{3}+\frac{\sqrt{10}}{3}\right)$.

38 c)

$$\begin{aligned} 3\sinh\theta - \cosh 2\theta &= 0 \\ 3\sinh\theta - (1+2\sinh^2\theta) &= 0 \\ -2\sinh^2\theta + 3\sinh\theta - 1 &= 0 \\ \sinh\theta = \frac{1}{2} \text{ or } \sinh\theta &= 1. \end{aligned}$$

If $\sinh\theta = \frac{1}{2}$, then $\theta = \ln\left(\frac{1}{2}+\sqrt{(\frac{1}{2})^2+1}\right) = \ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)$. If $\sinh\theta = 1$, then $\theta = \ln(1+\sqrt{1^2+1}) = \ln(1+\sqrt{2})$. So $\theta = \ln\left(\frac{1}{2}+\frac{\sqrt{5}}{2}\right)$ or $\theta = \ln(1+\sqrt{2})$.

38 d)

$$\begin{aligned} \cosh 2\theta - 4\cosh\theta &= 0 \\ (2\cosh^2\theta - 1) - 4\cosh\theta &= 0 \\ 2\cosh^2\theta - 4\cosh\theta - 1 &= 0 \\ \cosh\theta &= \frac{2\pm\sqrt{6}}{2}. \end{aligned}$$

If $\cosh\theta = \frac{2+\sqrt{6}}{2}$, then $\theta = \pm\ln\left(\frac{2+\sqrt{6}}{2}+\sqrt{(\frac{2+\sqrt{6}}{2})^2-1}\right) = \pm\ln\left(\frac{2+\sqrt{6}}{2}+\frac{\sqrt{6+4\sqrt{6}}}{2}\right)$. So $\theta = \ln\left(\frac{2+\sqrt{6}}{2}+\frac{\sqrt{6+4\sqrt{6}}}{2}\right)$ or $\theta = \ln\left(\frac{2+\sqrt{6}}{2}-\frac{\sqrt{6+4\sqrt{6}}}{2}\right)$.

39 a)

$$\begin{aligned}
\cosh 2\theta + 2\sinh\theta - 5 &= 0 \\
(1 + 2\sinh^2\theta) + 2\sinh\theta - 5 &= 0 \\
\sinh^2\theta + \sinh\theta - 2 &= 0 \\
(\sinh\theta + 2)(\sinh\theta - 1) &= 0 \\
\sinh\theta = -2 \text{ or } \sinh\theta &= 1.
\end{aligned}$$

If $\sinh\theta = -2$, then $\theta = \ln(-2 + \sqrt{(-2)^2 + 1}) = \ln(-2 + \sqrt{5})$. If $\sinh\theta = 1$, then $\theta = \ln(1 + \sqrt{1^2 + 1}) = \ln(1 + \sqrt{2})$. So $\theta = \ln(-2 + \sqrt{5})$ or $\theta = \ln(1 + \sqrt{2})$.

39 b)

$$\begin{aligned}
5\cosh 2\theta - 3\sinh\theta &= 5 \\
5(1 + 2\sinh^2\theta) - 3\sinh\theta &= 5 \\
10\sinh^2\theta - 3\sinh\theta &= 0 \\
\sinh\theta(10\sinh\theta - 3) &= 0 \\
\sinh\theta = 0 \text{ or } \sinh\theta &= \frac{3}{10}.
\end{aligned}$$

If $\sinh\theta = 0$, then $\theta = \ln(0 + \sqrt{0^2 + 1}) = \ln 1 = 0$. If $\sinh\theta = \frac{3}{10}$, then $\theta = \ln\left(\frac{3}{10} + \sqrt{(\frac{3}{10})^2 + 1}\right) = \ln\left(\frac{3}{10} + \frac{\sqrt{109}}{10}\right)$. So $\theta = 0$ or $\theta = \ln\left(\frac{3}{10} + \frac{\sqrt{109}}{10}\right)$.

39 c)

$$\begin{aligned}
8\cosh\theta - 3\cosh 2\theta &= 3 \\
8\cosh\theta - 3(2\cosh^2\theta - 1) - 3 &= 0 \\
-6\cosh^2\theta + 8\cosh\theta &= 0 \\
3\cosh^2\theta - 4\cosh\theta &= 0 \\
\cosh\theta(3\cosh\theta - 4) &= 0 \\
\cosh\theta = 0 \text{ or } \cosh\theta &= \frac{4}{3}.
\end{aligned}$$

$\cosh\theta = 0$ has no real solutions. If $\cosh\theta = \frac{4}{3}$, then $\theta = \pm\ln\left(\frac{4}{3} + \sqrt{(\frac{4}{3})^2 - 1}\right) = \pm\ln\left(\frac{4}{3} + \frac{\sqrt{7}}{3}\right)$. So $\theta = \ln\left(\frac{4}{3} + \frac{\sqrt{7}}{3}\right)$ or $\theta = \ln\left(\frac{4}{3} - \frac{\sqrt{7}}{3}\right)$.

39 d)

$$\begin{aligned}
5\cosh\theta - 12\cosh 2\theta + 17 &= 0 \\
5\cosh\theta - 12(2\cosh^2\theta - 1) + 17 &= 0 \\
5\cosh\theta - 24\cosh^2\theta + 12 + 17 &= 0 \\
24\cosh^2\theta - 5\cosh\theta - 29 &= 0 \\
(24\cosh\theta - 29)(\cosh\theta + 1) &= 0
\end{aligned}$$

$$\cosh\theta = \frac{29}{24} \text{ or } \cosh\theta = -1.$$

If $\cosh\theta = \frac{29}{24}$, then $\theta = \pm\ln\left(\frac{29}{24} + \sqrt{(\frac{29}{24})^2 - 1}\right) = \pm\ln\left(\frac{29}{24} + \frac{\sqrt{265}}{24}\right)$. $\cosh\theta = -1$ has no real solutions. So $\theta = \ln\left(\frac{29}{24} + \frac{\sqrt{265}}{24}\right)$ or $\theta = \ln\left(\frac{29}{24} - \frac{\sqrt{265}}{24}\right)$.

40 a)

$$\begin{aligned} 2\sinh 2\theta\,\mathrm{cosech}\,\theta &= 5 \\ 2\times 2\sinh\theta\cosh\theta\times\frac{1}{\sinh\theta} &= 5 \\ 4\cosh\theta &= 5 \\ \cosh\theta &= \frac{5}{4} \\ \theta &= \pm\ln\left(\frac{5}{4} + \sqrt{(\frac{5}{4})^2 - 1}\right) = \pm\ln\left(\frac{5}{4} + \frac{3}{4}\right) = \pm\ln 2. \end{aligned}$$

So $\theta = \pm\ln 2$.

40 b)

$$\begin{aligned} 5\sinh 2\theta\,\mathrm{sech}\,\theta &= 7 \\ 5\times 2\sinh\theta\cosh\theta\times\frac{1}{\cosh\theta} &= 7 \\ 10\sinh\theta &= 7 \\ \sinh\theta &= \frac{7}{10} \\ \theta &= \ln\left(\frac{7}{10} + \sqrt{(\frac{7}{10})^2 + 1}\right) = \ln\left(\frac{7}{10} + \frac{\sqrt{149}}{10}\right). \end{aligned}$$

So $\theta = \ln\left(\frac{7}{10} + \frac{\sqrt{149}}{10}\right)$.

40 c)

$$\begin{aligned} \cosh 2\theta\,\mathrm{cosech}\,\theta &= 3 \\ (1 + 2\sinh^2\theta)\times\frac{1}{\sinh\theta} &= 3 \\ 1 + 2\sinh^2\theta &= 3\sinh\theta \\ 2\sinh^2\theta - 3\sinh\theta + 1 &= 0 \\ \sinh\theta &= \frac{1}{2} \text{ or } \sinh\theta = 1. \end{aligned}$$

If $\sinh\theta = \frac{1}{2}$, then $\theta = \ln\left(\frac{1}{2} + \sqrt{(\frac{1}{2})^2 + 1}\right) = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$. If $\sinh\theta = 1$, then $\theta = \ln(1 + \sqrt{1^2 + 1}) = \ln(1 + \sqrt{2})$. So $\theta = \ln\left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right)$ or $\theta = \ln(1 + \sqrt{2})$.

40 d)

$$\begin{aligned} 2\cosh 2\theta \operatorname{sech}\theta &= 7 \\ 2(2\cosh^2\theta - 1) \times \frac{1}{\cosh\theta} &= 7 \\ 2(2\cosh^2\theta - 1) &= 7\cosh\theta \\ 4\cosh^2\theta - 7\cosh\theta - 2 &= 0 \\ \cosh\theta = -\frac{1}{4} \text{ or } \cosh\theta &= 2. \end{aligned}$$

$\cosh\theta = -\frac{1}{4}$ has no real solutions. If $\cosh\theta = 2$, then $\theta = \pm\ln(2+\sqrt{2^2-1}) = \pm\ln(2+\sqrt{3})$. So $\theta = \ln(2+\sqrt{3})$ or $\theta = \ln(2-\sqrt{3})$.

41 a)

$$\begin{aligned} 5\tanh 2x - 9\tanh x &= 0 \\ 5 \times \frac{2\tanh x}{1+\tanh^2 x} - 9\tanh x &= 0 \\ 10\tanh x - 9\tanh x(1+\tanh^2 x) &= 0 \\ \tanh x(10 - 9(1+\tanh^2 x)) &= 0 \\ \tanh x(1 - 9\tanh^2 x) &= 0 \\ \tanh x = 0 \text{ or } \tanh^2 x = \frac{1}{9} &\Rightarrow \tanh x = \pm\frac{1}{3}. \end{aligned}$$

If $\tanh x = 0$, then $x = \frac{1}{2}\ln\left(\frac{1+0}{1-0}\right) = \frac{1}{2}\ln 1 = 0$. If $\tanh x = \frac{1}{3}$, then $x = \frac{1}{2}\ln\left(\frac{1+\frac{1}{3}}{1-\frac{1}{3}}\right) = \frac{1}{2}\ln 2$. If $\tanh x = -\frac{1}{3}$, then $x = \frac{1}{2}\ln\left(\frac{1+(-\frac{1}{3})}{1-(-\frac{1}{3})}\right) = \frac{1}{2}\ln\frac{1}{2} = -\frac{1}{2}\ln 2$. So $x = 0$, $x = \frac{1}{2}\ln 2$ or $x = -\frac{1}{2}\ln 2$.

41 b)

$$\begin{aligned} 4\tanh x - 3\tanh 2x &= 0 \\ 4\tanh x - 3 \times \frac{2\tanh x}{1+\tanh^2 x} &= 0 \\ 4\tanh x(1+\tanh^2 x) - 6\tanh x &= 0 \\ 2\tanh x(1+\tanh^2 x) - 3\tanh x &= 0 \\ \tanh x(2(1+\tanh^2 x) - 3) &= 0 \\ \tanh x(2\tanh^2 x - 1) &= 0 \\ \tanh x = 0 \text{ or } \tanh x = \pm\frac{1}{\sqrt{2}} &= \pm\frac{\sqrt{2}}{2}. \end{aligned}$$

If $\tanh x = 0$, then $x = 0$. If $\tanh x = \frac{\sqrt{2}}{2}$, then $x = \frac{1}{2}\ln\left(\frac{1+\frac{\sqrt{2}}{2}}{1-\frac{\sqrt{2}}{2}}\right) = \frac{1}{2}\ln(3+2\sqrt{2})$. If $\tanh x = -\frac{\sqrt{2}}{2}$, then $x = \frac{1}{2}\ln\left(\frac{1-\frac{\sqrt{2}}{2}}{1+\frac{\sqrt{2}}{2}}\right) = \frac{1}{2}\ln(3-2\sqrt{2})$. So $x = 0$, $x = \frac{1}{2}\ln(3+2\sqrt{2})$ or $x = \frac{1}{2}\ln(3-2\sqrt{2})$.

42)

$$\begin{aligned}\sinh x &= \frac{4}{3}\\ \cosh^2 x - \sinh^2 x = 1 \Rightarrow \cosh^2 x - \left(\frac{4}{3}\right)^2 &= 1\\ \cosh^2 x &= \frac{25}{9}\\ \cosh x &= \frac{5}{3} \text{ or } \cosh x = -\frac{5}{3}.\end{aligned}$$

$\cosh x = -\frac{5}{3}$ has no real solutions, so $\cosh x = \frac{5}{3}$ only. Therefore, $\sinh 2x = 2\sinh x \cosh x = 2 \times \frac{4}{3} \times \frac{5}{3} = \frac{40}{9}$.

43)

$$\sinh x = -\frac{1}{3} \quad \therefore \quad \cosh 2x = 1 + 2\sinh^2 x = 1 + 2 \times \left(-\frac{1}{3}\right)^2 = \frac{11}{9}.$$

44)

$$\begin{aligned}\cosh x &= \frac{9}{4}\\ \cosh^2 x - \sinh^2 x = 1 \Rightarrow \left(\frac{9}{4}\right)^2 - \sinh^2 x &= 1\\ \sinh^2 x &= \frac{65}{16}\\ \sinh x &= \frac{\sqrt{65}}{4} \text{ or } \sinh x = -\frac{\sqrt{65}}{4}.\end{aligned}$$

Thus $\tanh x = \frac{\sqrt{65}}{9}$ or $\tanh x = -\frac{\sqrt{65}}{9}$, and

$$\tanh 2x = \frac{2\tanh x}{1 + \tanh^2 x} = \frac{2 \times \frac{\sqrt{65}}{9}}{1 + (\frac{\sqrt{65}}{9})^2} = \frac{9\sqrt{65}}{73}$$

or $\tanh 2x = -\frac{9\sqrt{65}}{73}$. So $\tanh 2x = \pm\frac{9\sqrt{65}}{73}$.

45 a)

$$\begin{aligned}\sinh 3\theta &\equiv \sinh(2\theta + \theta)\\ &\equiv \sinh 2\theta \cosh\theta + \cosh 2\theta \sinh\theta\\ &\equiv (2\sinh\theta\cosh\theta)\cosh\theta + (1 + 2\sinh^2\theta)\sinh\theta\\ &\equiv 2\sinh\theta\cosh^2\theta + \sinh\theta + 2\sinh^3\theta\\ &\equiv 2\sinh\theta(1 + \sinh^2\theta) + \sinh\theta + 2\sinh^3\theta\\ &\equiv 2\sinh\theta + 2\sinh^3\theta + \sinh\theta + 2\sinh^3\theta\\ &\equiv 4\sinh^3\theta + 3\sinh\theta.\end{aligned}$$

45 b)

$$\begin{aligned}
20x^3 + 15x - 12 &= 0 \\
20\sinh^3\theta + 15\sinh\theta - 12 &= 0 \\
5(4\sinh^3\theta + 3\sinh\theta) - 12 &= 0 \\
5\sinh 3\theta - 12 &= 0 \\
\sinh 3\theta &= \frac{12}{5} \\
3\theta &= \ln\left(\frac{12}{5} + \sqrt{(\frac{12}{5})^2 + 1}\right) \\
\theta &= \frac{1}{3}\ln\left(\frac{12}{5} + \frac{13}{5}\right) = \frac{1}{3}\ln 5.
\end{aligned}$$

45 c)

$$\begin{aligned}
x = \sinh\theta &= \sinh\left(\frac{1}{3}\ln 5\right) \\
&= \frac{e^{\frac{1}{3}\ln 5} - e^{-\frac{1}{3}\ln 5}}{2} \\
&= \frac{e^{\ln 5^{1/3}} - e^{\ln 5^{-1/3}}}{2} \\
&= \frac{1}{2}(5^{1/3} - 5^{-1/3}).
\end{aligned}$$

46 a)

$$\begin{aligned}
\text{LHS: } \coth^2 x - \tanh^2 x &\equiv (\operatorname{cosech}^2 x + 1) - (1 - \operatorname{sech}^2 x) \\
&\equiv \operatorname{cosech}^2 x + \operatorname{sech}^2 x \quad : \text{RHS.}
\end{aligned}$$

46 b)

$$\begin{aligned}
\text{RHS: } \sinh^2 x - \tanh^2 x &\equiv \sinh^2 x - \frac{\sinh^2 x}{\cosh^2 x} \\
&\equiv \sinh^2 x\left(1 - \frac{1}{\cosh^2 x}\right) \\
&\equiv \sinh^2 x(1 - \operatorname{sech}^2 x) \\
&\equiv \sinh^2 x \tanh^2 x \quad : \text{LHS.}
\end{aligned}$$

46 c)

$$\text{LHS: } \operatorname{cosech}^2 x - \operatorname{sech}^2 x \equiv \frac{1}{\sinh^2 x} - \frac{1}{\cosh^2 x}$$

$$
\begin{aligned}
&\equiv \frac{\cosh^2 x - \sinh^2 x}{\sinh^2 x \cosh^2 x} \\
&\equiv \frac{1}{\sinh^2 x \cosh^2 x} \\
&\equiv \frac{1}{(\sinh x \cosh x)^2} \\
&\equiv \frac{4}{(2 \sinh x \cosh x)^2} \\
&\equiv \frac{4}{\sinh^2 2x} \\
&\equiv 4 \operatorname{cosech}^2 2x \quad : \text{RHS.}
\end{aligned}
$$

47 a)

$$
\begin{aligned}
\text{LHS: } (\operatorname{sech} x - \tanh x)^2 &\equiv \operatorname{sech}^2 x - 2 \operatorname{sech} x \tanh x + \tanh^2 x \\
&\equiv (1 - \tanh^2 x) - 2 \operatorname{sech} x \tanh x + \tanh^2 x \\
&\equiv 1 - 2 \operatorname{sech} x \tanh x \quad : \text{RHS.}
\end{aligned}
$$

47 b)

$$
\begin{aligned}
\text{LHS: } (\operatorname{sech} x - \coth x)^2 &\equiv \operatorname{sech}^2 x - 2 \operatorname{sech} x \coth x + \coth^2 x \\
&\equiv (1 - \tanh^2 x) - 2 \times \frac{1}{\cosh x} \times \frac{\cosh x}{\sinh x} + (\operatorname{cosech}^2 x + 1) \\
&\equiv 1 - \tanh^2 x - 2 \times \frac{1}{\sinh x} + \operatorname{cosech}^2 x + 1 \\
&\equiv 2 - \tanh^2 x - 2 \operatorname{cosech} x + \operatorname{cosech}^2 x \quad : \text{RHS.}
\end{aligned}
$$

47 c)

$$
\begin{aligned}
\text{LHS: } (\operatorname{sech} x - \operatorname{cosech} x)^2 &\equiv \left(\frac{1}{\cosh x} - \frac{1}{\sinh x} \right)^2 \\
&\equiv \left(\frac{\sinh x - \cosh x}{\sinh x \cosh x} \right)^2 \\
&\equiv \frac{(\sinh x - \cosh x)^2}{\sinh^2 x \cosh^2 x} \\
&\equiv \operatorname{cosech}^2 x \operatorname{sech}^2 x (\sinh x - \cosh x)^2 \\
&\equiv \operatorname{cosech}^2 x \operatorname{sech}^2 x (\cosh^2 x + \sinh^2 x - 2 \sinh x \cosh x) \\
&\equiv \operatorname{cosech}^2 x \operatorname{sech}^2 x (\cosh 2x - \sinh 2x) \quad : \text{RHS.}
\end{aligned}
$$

48 a)

$$
\text{LHS: } \frac{1 + \cosh x}{\sinh x} - \frac{\sinh x}{1 + \cosh x} \equiv \frac{(1 + \cosh x)^2 - \sinh^2 x}{\sinh x (1 + \cosh x)}
$$

$$
\begin{aligned}
&\equiv \frac{1+2\cosh x+\cosh^2 x-\sinh^2 x}{\sinh x(1+\cosh x)}\\
&\equiv \frac{1+2\cosh x+1}{\sinh x(1+\cosh x)}\\
&\equiv \frac{2(1+\cosh x)}{\sinh x(1+\cosh x)}\\
&\equiv \frac{2}{\sinh x}\\
&\equiv 2\operatorname{cosech} x \quad : \text{RHS.}
\end{aligned}
$$

48 b)

$$
\begin{aligned}
\frac{1+\cosh t}{\sinh t}-\frac{\sinh t}{1+\cosh t} &= -4\\
2\operatorname{cosech} t &= -4\\
\operatorname{cosech} t &= -2\\
\sinh t &= -\frac{1}{2}\\
t &= \ln\left(-\frac{1}{2}+\sqrt{(-\frac{1}{2})^2+1}\right) = \ln\left(-\frac{1}{2}+\frac{\sqrt{5}}{2}\right).
\end{aligned}
$$

49 a)

$$
\begin{aligned}
\text{LHS: } \frac{\sinh x}{\cosh x-1}-\operatorname{cosech} x &\equiv \frac{\sinh x}{\cosh x-1}\times\frac{\cosh x+1}{\cosh x+1}-\frac{1}{\sinh x}\\
&\equiv \frac{\sinh x\cosh x+\sinh x}{\cosh^2 x-1}-\frac{1}{\sinh x}\\
&\equiv \frac{\sinh x\cosh x+\sinh x}{\sinh^2 x}-\frac{\sinh x}{\sinh^2 x}\\
&\equiv \frac{\sinh x\cosh x+\sinh x-\sinh x}{\sinh^2 x}\\
&\equiv \frac{\sinh x\cosh x}{\sinh^2 x}\\
&\equiv \frac{\cosh x}{\sinh x}\\
&\equiv \coth x \quad : \text{RHS.}
\end{aligned}
$$

49 b)

$$
\begin{aligned}
\frac{\sinh t}{\cosh t-1} &= 8+\coth t\\
\coth t+\operatorname{cosech} t &= 8+\coth t\\
\operatorname{cosech} t &= 8\\
\sinh t &= \frac{1}{8}\\
t &= \ln\left(\frac{1}{8}+\sqrt{(\frac{1}{8})^2+1}\right) = \ln\left(\frac{1}{8}+\frac{\sqrt{65}}{8}\right).
\end{aligned}
$$

50)

$$\begin{aligned}
\operatorname{sech}\frac{x}{2} &= 6\sinh\frac{x}{2} \\
\frac{1}{\cosh\frac{x}{2}} &= 6\sinh\frac{x}{2} \\
1 &= 6\sinh\frac{x}{2}\cosh\frac{x}{2} \\
1 &= 3\sinh x \\
\sinh x &= \frac{1}{3} \\
x &= \ln\left(\frac{1}{3}+\sqrt{(\frac{1}{3})^2+1}\right) = \ln\left(\frac{1+\sqrt{10}}{3}\right) \\
&= \ln(1+\sqrt{10}) - \ln 3 \quad \text{as required.}
\end{aligned}$$

51 a)

$$\begin{aligned}
\text{LHS: } \coth 2x - \tanh 2x &\equiv \frac{1}{\tanh 2x} - \tanh 2x \\
&\equiv \frac{1-\tanh^2 2x}{\tanh 2x} \\
&\equiv \frac{\operatorname{sech}^2 2x}{\tanh 2x} \\
&\equiv \frac{1}{\cosh^2 2x}\frac{\sinh 2x}{\cosh 2x} \\
&\equiv \frac{1}{\cosh^2 2x} \times \frac{\cosh 2x}{\sinh 2x} \\
&\equiv \frac{1}{\cosh 2x \sinh 2x} \\
&\equiv \frac{2}{2\sinh 2x \cosh 2x} \\
&\equiv \frac{2}{\sinh 4x} \\
&\equiv 2\operatorname{cosech} 4x \quad \text{: RHS.}
\end{aligned}$$

51 b)

$$\begin{aligned}
\text{LHS: } \coth x - \tanh 2x &\equiv \frac{1}{\tanh x} - \frac{2\tanh x}{1+\tanh^2 x} \\
&\equiv \frac{1+\tanh^2 x - 2\tanh^2 x}{\tanh x(1+\tanh^2 x)} \\
&\equiv \frac{1-\tanh^2 x}{\tanh x(1+\tanh^2 x)} \\
&\equiv \frac{\operatorname{sech}^2 x}{\tanh x(1+\tanh^2 x)} \\
&\equiv \frac{1/\cosh^2 x}{(\sinh x/\cosh x)(1+\sinh^2 x/\cosh^2 x)}
\end{aligned}$$

$$
\begin{aligned}
&\equiv \frac{1/\cosh^2 x}{(\sinh x/\cosh x)(1+\sinh^2 x/\cosh^2 x)} \times \frac{\cosh^2 x}{\cosh^2 x} \\
&\equiv \frac{1}{\sinh x \cosh x(\cosh^2 x + \sinh^2 x)} \\
&\equiv \frac{\cosh x}{\sinh x} \times \frac{1}{\cosh^2 x + \sinh^2 x} \\
&\equiv \coth x \times \frac{1}{\cosh 2x} \\
&\equiv \coth x \operatorname{sech} 2x \quad : \text{RHS.}
\end{aligned}
$$

52 a)

$$
\begin{aligned}
\text{LHS: } \coth 2x - \cosh 2x &\equiv \frac{\cosh 2x}{\sinh 2x} - \cosh 2x \\
&\equiv \cosh 2x \left(\frac{1}{\sinh 2x} - 1\right) \\
&\equiv \cosh 2x \left(\frac{1-\sinh 2x}{\sinh 2x}\right) \\
&\equiv \coth 2x(1-\sinh 2x) \quad : \text{RHS.}
\end{aligned}
$$

52 b)

$$
\begin{aligned}
\text{LHS: } \coth 2x + \operatorname{cosech} 2x &\equiv \frac{\cosh 2x}{\sinh 2x} + \frac{1}{\sinh 2x} \\
&\equiv \frac{\cosh 2x + 1}{\sinh 2x} \\
&\equiv \frac{2\cosh^2 x - 1 + 1}{2\sinh x \cosh x} \\
&\equiv \frac{2\cosh^2 x}{2\sinh x \cosh x} \\
&\equiv \frac{\cosh x}{\sinh x} \\
&\equiv \coth x \quad : \text{RHS.}
\end{aligned}
$$

53)

$$
\begin{aligned}
\text{RHS: } \tanh 2x + \operatorname{cosech} 2x &\equiv \frac{\sinh 2x}{\cosh 2x} + \frac{1}{\sinh 2x} \\
&\equiv \frac{\sinh^2 2x + \cosh 2x}{\sinh 2x \cosh 2x} \\
&\equiv \frac{1}{\sinh 2x \cosh 2x}(\sinh^2 2x + \cosh 2x) \\
&\equiv \frac{2}{2\sinh 2x \cosh 2x}(\sinh^2 2x + \cosh 2x) \\
&\equiv \frac{2}{\sinh 4x}(\sinh^2 2x + \cosh 2x) \\
&\equiv 2\operatorname{cosech} 4x(\sinh^2 2x + \cosh 2x) \quad : \text{LHS.}
\end{aligned}
$$

54 a)

$$\begin{aligned}\text{LHS: } (\sinh 2x + \cosh 2x)^2 &\equiv \sinh^2 2x + 2\sinh 2x \cosh 2x + \cosh^2 2x \\ &\equiv 2\sinh 2x \cosh 2x + (\cosh^2 2x + \sinh^2 2x) \\ &\equiv \sinh 4x + \cosh 4x \quad \text{: RHS.}\end{aligned}$$

54 b)

$$\begin{aligned}\text{LHS: } \frac{2\sinh x}{\cosh 2x - 1} - \frac{2\sinh x}{\cosh 2x + 1} & \\ &\equiv \frac{2\sinh x(\cosh 2x + 1) - 2\sinh x(\cosh 2x - 1)}{(\cosh 2x - 1)(\cosh 2x + 1)} \\ &\equiv \frac{2\sinh x \cosh 2x + 2\sinh x - 2\sinh x \cosh 2x + 2\sinh x}{\cosh^2 2x - 1} \\ &\equiv \frac{4\sinh x}{\cosh^2 2x - 1} \\ &\equiv \frac{4\sinh x}{\sinh^2 2x} \\ &\equiv \frac{4\sinh x}{(2\sinh x \cosh x)^2} \\ &\equiv \frac{4\sinh x}{4\sinh^2 x \cosh^2 x} \\ &\equiv \frac{1}{\sinh x \cosh^2 x} \\ &\equiv \operatorname{cosech} x \operatorname{sech}^2 x \quad \text{: RHS.}\end{aligned}$$

55 a)

$$\begin{aligned}\text{RHS: } \frac{1}{(\sinh x + \cosh x)^2} &\equiv \frac{1}{(\sinh x + \cosh x)^2} \times \frac{(\sinh x - \cosh x)^2}{(\sinh x - \cosh x)^2} \\ &\equiv \frac{(\sinh x - \cosh x)^2}{(\sinh x + \cosh x)(\sinh x - \cosh x)(\sinh x + \cosh x)(\sinh x - \cosh x)} \\ &\equiv \frac{(\sinh x - \cosh x)^2}{(\sinh^2 x - \cosh^2 x)(\sinh^2 x - \cosh^2 x)} \\ &\equiv \frac{(\sinh x - \cosh x)^2}{1 \times 1} \\ &\equiv (\sinh x - \cosh x)^2 \\ &\equiv \sinh^2 x - 2\sinh x \cosh x + \cosh^2 x \\ &\equiv (\cosh^2 x + \sinh^2 x) - 2\sinh x \cosh x \\ &\equiv \cosh 2x - \sinh 2x \quad \text{: LHS.}\end{aligned}$$

55 b)

$$\text{RHS: } \frac{\cosh x(\coth x - 1)}{\sinh x + \cosh x} \equiv \frac{\cosh x(\frac{\cosh x}{\sinh x} - 1)}{\sinh x + \cosh x}$$

$$
\begin{aligned}
&\equiv \frac{\cosh x}{\sinh x}\frac{\cosh x - \sinh x}{\sinh x + \cosh x} \\
&\equiv \frac{\cosh x}{\sinh x} \times \frac{\cosh x - \sinh x}{\cosh x + \sinh x} \times \frac{\cosh x - \sinh x}{\cosh x - \sinh x} \\
&\equiv \frac{\cosh x}{\sinh x} \times \frac{\cosh^2 x - 2\sinh x \cosh x + \sinh^2 x}{\cosh^2 x - \sinh^2 x} \\
&\equiv \frac{\cosh x}{\sinh x} \times \frac{(\cosh^2 x + \sinh^2 x) - 2\sinh x \cosh x}{1} \\
&\equiv \coth x \times \frac{\cosh 2x - \sinh 2x}{1} \\
&\equiv \coth x(\cosh 2x - \sinh 2x) \quad : \text{LHS.}
\end{aligned}
$$

56)

$$
\begin{aligned}
6\sinh x \cosh x - 8\sinh x - 15\cosh x + 20 &= 0 \\
(2\sinh x - 5)(3\cosh x - 4) &= 0 \\
\sinh x = \frac{5}{2} \text{ or } \cosh x &= \frac{4}{3}.
\end{aligned}
$$

If $\sinh x = \frac{5}{2}$, then

$$x = \ln\left(\frac{5}{2} + \sqrt{(\frac{5}{2})^2 + 1}\right) = \ln\left(\frac{5}{2} + \frac{\sqrt{29}}{2}\right).$$

If $\cosh x = \frac{4}{3}$, then

$$x = \pm\ln\left(\frac{4}{3} + \sqrt{(\frac{4}{3})^2 - 1}\right) = \pm\ln\left(\frac{4}{3} + \frac{\sqrt{7}}{3}\right).$$

So $x = \ln\left(\frac{5}{2} + \frac{\sqrt{29}}{2}\right)$, $x = \ln\left(\frac{4}{3} + \frac{\sqrt{7}}{3}\right)$, or $\ln\left(\frac{4}{3} - \frac{\sqrt{7}}{3}\right)$.

57 a)

$$
\begin{aligned}
\sinh(\ln(8 + 2x)) &= x \\
\ln(8 + 2x) &= \operatorname{arsinh} x \\
\ln(8 + 2x) &= \ln(x + \sqrt{x^2 + 1}) \\
8 + 2x &= x + \sqrt{x^2 + 1} \\
8 + x &= \sqrt{x^2 + 1} \\
(8 + x)^2 &= x^2 + 1 \\
x^2 + 16x + 64 &= x^2 + 1 \\
16x &= -63 \\
x &= -\frac{63}{16}.
\end{aligned}
$$

57 b)

$$\cosh(\ln(3 - 5x)) + 10x = 0$$

$$\begin{aligned}
\cosh(\ln(3-5x)) &= -10x \\
\ln(3-5x) &= \operatorname{arcosh}(-10x) \\
\ln(3-5x) &= \ln\left(-10x+\sqrt{(-10x)^2-1}\right) \\
3-5x &= -10x+\sqrt{100x^2-1} \\
3+5x &= \sqrt{100x^2-1} \\
(3+5x)^2 &= 100x^2-1 \\
25x^2+30x+9 &= 100x^2-1 \\
-75x^2+30x+10 &= 0 \\
15x^2-6x-2 &= 0 \\
x &= \frac{3\pm\sqrt{39}}{15}.
\end{aligned}$$

However, $x=\frac{3+\sqrt{39}}{15}$ cannot be a solution as $3-5(\frac{3+\sqrt{39}}{15})=\frac{6-\sqrt{39}}{3}<0$ and we cannot take the logarithm of a negative number. So $x=\frac{3-\sqrt{39}}{15}$ is the only real solution.

57 c)

$$\begin{aligned}
\operatorname{sech}(\ln(4x+9))+\frac{1}{12x} &= 0 \\
\operatorname{sech}(\ln(4x+9)) &= -\frac{1}{12x} \\
\cosh(\ln(4x+9)) &= -12x \\
\ln(4x+9) &= \operatorname{arcosh}(-12x) \\
\ln(4x+9) &= \ln\left(-12x+\sqrt{(-12x)^2-1}\right) \\
4x+9 &= -12x+\sqrt{144x^2-1} \\
16x+9 &= \sqrt{144x^2-1} \\
(16x+9)^2 &= 144x^2-1 \\
256x^2+288x+81 &= 144x^2-1 \\
112x^2+288x+82 &= 0 \\
x &= \frac{-36\pm19\sqrt{2}}{28}.
\end{aligned}$$

57 d)

$$\begin{aligned}
\tanh(\ln(\sqrt{3}+4x)) &= 2x \\
\ln(\sqrt{3}+4x) &= \operatorname{artanh}(2x) \\
\ln(\sqrt{3}+4x) &= \frac{1}{2}\ln\left(\frac{1+2x}{1-2x}\right) \\
\frac{1}{2}\ln(3+4x) &= \frac{1}{2}\ln\left(\frac{1+2x}{1-2x}\right) \\
\ln(3+4x) &= \ln\left(\frac{1+2x}{1-2x}\right)
\end{aligned}$$

$$
\begin{aligned}
3 + 4x &= \frac{1 + 2x}{1 - 2x} \\
(3 + 4x)(1 - 2x) &= 1 + 2x \\
-8x^2 - 2x + 3 &= 1 + 2x \\
-8x^2 - 4x + 2 &= 0 \\
4x^2 + 2x - 1 &= 0 \\
x &= \frac{-1 \pm \sqrt{5}}{4}.
\end{aligned}
$$

However, $x = \frac{-1-\sqrt{5}}{4}$ cannot be a solution as $3 + 4(\frac{-1-\sqrt{5}}{4}) = 2 - \sqrt{5} < 0$ and we cannot take the square root of a negative number. So $x = \frac{-1+\sqrt{5}}{4}$ is the only real solution.